Shadows, Fibres and Secants of the Posterior Tower

Conditional expectations, Beck defects, and positive tropicalisation

Vinay K Awasthi
Department of Applied Mathematics and Statistics, Johns Hopkins University
vawasth1@jh.edu

October 13, 2026

Abstract

The three-level posterior tower [1] has two candidate mixed distributive laws, and both fail some Beck axioms. This paper studies the failures through two operations on the tower's levels: *pooling*, which replaces a law by conditional expectations of it, and *tropicalisation*, which replaces probabilities by their large-deviation rates. It unifies and corrects three earlier manuscripts, on the Giry-monad formulation, on conditional shadows, and on fibres and secants. The changes are listed in appendix A.

Algebra. For finite partitions A, B of a full-support probability space, five conditions are equivalent:

- $E_A E_B = E_B E_A$;
- $E_A E_B = E_{A \wedge B}$;
- A and B are conditionally independent given the meet $A \wedge B$;
- every connected component of the bipartite cell graph has a rank-one contingency block;
- every singular value of the normalised contingency matrix is 0 or 1.

The identity $E_A E_B = E_{A \vee B}$ holds only when $A = B$. In general $(E_A E_B)^n \to E_{A \wedge B}$ at the exact rate c^{2n-1}, so interchange holds laxly, through the meet.

Shadows. A coherent context determines a star-shaped probability space centred at the level-one predictive. Shadows are conditional expectations on this fixed space. They form a commutative band and satisfy interchange strictly. Shadows defined by label maps on laws do not compose as conditional expectations.

Topology and order. The barycentre has two continuous splittings, $e_\perp = \eta\mu$ and $e_\top = \mathsf{D}(\eta)\mu$. These form a left-zero band and bound every barycentre fibre in convex order. A coherent context is a three-step simplex-valued martingale. The shift law's level-one unit defect is the full convex-order diameter of a fibre; its multiplication defect is one conditional-expectation step.

Categorical structure. We construct the lifting functor from contexts to measure spaces in two versions and prove naturality of the shadows along measure-preserving maps. The double category is the quintet construction of a locally preordered 2-category, in which interchange is automatic. In this language each Beck defect computed so far is a single conditional-expectation step. For coupling liftings of Barr and martingale type there is a no-go theorem: the comultiplication cell of the averaging law is oriented only when labels are pooled away, which is incompatible with the counit and the comultiplication.

Geometry. Over a real closed non-Archimedean field the valuation is a monad morphism to the min-plus distribution monad. Classical barycentre fibres map onto tropical fibres. Positive mixtures map exactly onto tropical secants. The valuation factors through the Berkovich analytification, and on positive points the square commutes. The Bayes posterior tropicalises to the

hard assignment, so the Voronoi walls of nearest-atom pooling are tropical walls. We also correct the Gaussian fusion square, the finite stochastic-to-convex functor and the moment-variety programme.

Cohomology. Conditional expectations are the restriction maps of the L^2 presheaf on the lattice of σ-algebras. Its Beck–Chevalley condition is exactly conditional independence given the meet, with defect norm c. Compatible sections always glue in finite dimensions, and gluing fails only when $c = 1$. The Beck comparisons are restriction maps, never isomorphisms. Tropical fibres are covered by assignment cells. The nerve computes their cohomology, and Farkas certificates of impossible assignment patterns, the tropical form of the tower's lifting certificates, determine it. For Hardy–Weinberg mixtures the fibre has $b_0 = 2$ off the collision locus and $b_0 = 1$ on it, with $b_1 = 0$, so the collision wall is a jump in H^0.

Hallucination. In a three-token model observed through pairwise windows, a hallucination (an obstructed claim) is a failure of gluing. It occurs below a straight wall in source-weight space, at agreement $\frac{1}{3}$, and is certified by a Bell-type functional. Above the wall, the two overlapping windows satisfy Beck–Chevalley iff the tokens are Markov through the shared token. The defect is $c = |2\varepsilon - 1|/(2\varepsilon)$ at the symmetric section. Tropicalisation sees only support-level hallucinations. A pre-registered test on public GPT-3.5 and GPT-4 answers finds that cross-run disagreement detects wrong answers in all five tasks (AUROC 0.69–0.88), and that the window defect exceeds its Markov null in three of them. One worked example, the Hardy–Weinberg atlas, illustrates every layer in seven figures generated by a script.

Places. Read over the Berkovich spectrum $\mathcal{M}(\mathbb{Z})$, the context presheaf separates by place: the window data always glue over $\mathbb{Q}$ and every $\mathbb{Q}_p$, Beck–Chevalley is decided at the trivial seminorm, and the hallucination obstruction is supported at the archimedean place, the only one that carries order. A second study, on HaluEval with a local model, finds that the disagreement detector adds information beyond answer length and that the within-sample windows fail Beck–Chevalley through item-level difficulty. A companion note applies the same tools to pooling several agents. Nine scripts check every finite claim (30, 8, 10, 9, 9, 4, 5 and 12 checks, and 7 Monte Carlo checks of the statistical test), all passing. Code, figures and an interactive explorer are at `https://github.com/vkawasth/Trainingless-Transformer-NoGradientDescent/tree/main/papers`.

Contents

1 Introduction **4**
- 1.1 The problem . . . 4
- 1.2 Guiding principles . . . 5
- 1.3 Main results . . . 5
- 1.4 Status . . . 6
- 1.5 Conventions . . . 6

2 The posterior tower **6**
- 2.1 Levels and contexts . . . 6
- 2.2 The absolute tower and the two laws . . . 7
- 2.3 Beck axioms . . . 7

3 Algebra of conditional expectations **7**
- 3.1 The partition lattice . . . 7
- 3.2 The commutation theorem . . . 7
- 3.3 Worked regimes . . . 8
- 3.4 The general case: rate and lax interchange . . . 9
- 3.5 Conditioning on a third variable . . . 9

4 Shadows **10**
4.1 Label-defined shadows on laws . . . 10
4.2 Labelled random measures . . . 10
4.3 The star space of a context and the canonical shadows . . . 11

5 Topology and order: how the tower acts **11**
5.1 Two idempotents . . . 11
5.2 Barycentre fibres . . . 12
5.3 A coherent context is a martingale . . . 12
5.4 The Beck defects located . . . 13
5.5 Continuity . . . 13

6 Categorical structure **13**
6.1 Why a separate construction is needed . . . 13
6.2 The context category . . . 13
6.3 The lifting functor, in two versions . . . 14
6.4 Shadows along context maps . . . 14
6.5 Typed squares and quintets . . . 15
6.6 What the tower supplies, and what it lacks . . . 15
6.7 Which lifting? Three facts that sharpen the problem . . . 16
6.8 The comultiplication and a no-go theorem . . . 17
6.9 Escaping thinness: the coupling 2-category . . . 18

7 Beck defects as comparison cells **19**
7.1 What a factorisation requires . . . 19
7.2 One-shadow and two-shadow options . . . 20
7.3 Higher cells . . . 21

8 The measure-theoretic (Giry) extension **21**

9 The finite stochastic-to-convex functor **22**

10 A Gaussian model: fusion and its type-correct domain **22**

11 Positive tropicalisation and the Berkovich factorisation **23**
11.1 Why a real non-Archimedean field . . . 23
11.2 The tropical tower . . . 23
11.3 Fibres . . . 24
11.4 Tropical secants . . . 24
11.5 The Berkovich square . . . 25
11.6 What the analogies do not give . . . 25

12 Shadows at zero temperature and secant geometry **26**
12.1 The posterior tropicalises to the hard assignment . . . 26
12.2 Statistical models as varieties . . . 26
12.3 Walls of a component collision . . . 27

13 The Hardy–Weinberg atlas **27**

14 Cohomology I: the L^2 presheaf and its Beck–Chevalley condition **30**
14.1 Gluing and the degree-one obstruction . . . 32
14.2 A unified coefficient system . . . 32

15 Cohomology II: fibres as topology **34**
15.1 The cover by assignment cells . . . 35
15.2 Local lifts as sections . . . 35
15.3 The computation for Hardy–Weinberg . . . 36

16 Example: hallucination in a three-token model **37**
16.1 The model . . . 38
16.2 Hallucination is a failure of gluing . . . 38
16.3 The wall in weight space, and a token stream . . . 38
16.4 The L^2 presheaf on context windows . . . 39
16.5 Three layers of obstruction . . . 40
16.6 What tropicalisation sees . . . 41

17 Places: the context presheaf over $\mathcal{M}(\mathbb{Z})$ **41**

18 Empirical tests of the hallucination example **46**
18.1 Why only the second layer can be tested on existing data . . . 46
18.2 Study 1: LLM-drift answers . . . 46
18.3 Study 2: HaluEval QA . . . 47
18.4 Study 3: kit . . . 49

19 Computational checks **49**
19.1 The scripts . . . 49
19.2 What checks establish . . . 50
19.3 Audit of earlier scripts . . . 50

20 Consequence for the tower paper **51**

21 Summary and open problems **51**

A Changes from the previous versions **52**

Code and data availability **53**

Acknowledgements **53**

1 Introduction

1.1 The problem

A three-level posterior tower records three things: a predictive $\rho \in \mathsf{D}X$, a law $\Pi \in \mathsf{D}^2X$ over predictives, and a law $\Omega \in \mathsf{D}^3X$ over such laws. It also carries a shadow Σ of labelled hypotheses. A candidate mixed distributive law on the tower may fail some Beck axioms, and the failures are comparisons between constructions, not merely numerical discrepancies. The motivating question is whether two ways of pooling information organise these comparisons, and whether the comparisons have a geometric meaning. Three logically separate tasks are involved:

(1) define the pooling operators (shadows) and prove their operator identities;

(2) build a category or double category in which the Beck comparisons are well-typed cells;

(3) locate the comparisons geometrically: in barycentre fibres, in convex order, and after tropicalisation, in secant geometry.

None of these implies another, and the paper keeps them apart.

1.2 Guiding principles

(i) *Compose operators on a fixed probability space.* Composing measure-valued maps that recompute their partitions after each step is a different operation. Most earlier difficulties came from mixing the two.

(ii) *Conditional expectations interact through the meet, never the join.*

(iii) *Probability tropicalises cleanly because it is positive.* Leading terms of positive series cannot cancel, so the valuation is a semiring homomorphism. Every geometric statement in the paper rests on this.

1.3 Main results

(A) **Commutation** (theorem 3.3): five equivalent conditions, a spectral and a combinatorial criterion among them. The join identity holds only for equal σ-algebras (proposition 3.4).

(B) **Rate and lax interchange** (theorem 3.6 and corollary 3.8).

(C) **Shadow algebra** (theorem 4.9): the canonical shadows of a context form a commutative band.

(D) **Fibres and martingale** (propositions 5.1 and 5.2 and theorem 5.3).

(E) **Beck defects located** (propositions 5.4 and 7.2).

(F) **Lifting functor and naturality** (propositions 4.6, 6.5 and 6.6); quintet double category (proposition 6.10).

(G) **Tropical tower** (theorem 11.3), **fibre lifting** (theorem 11.6), **tropical secants** (theorem 11.9) and the **Berkovich square** (theorem 11.11).

(H) **Zero-temperature shadows and walls** (propositions 12.1 and 12.2).

(I) **Cohomology**: Beck–Chevalley for the L^2 presheaf is conditional independence (theorem 14.4); gluing and the degree-one obstruction (theorem 14.6); fibre cohomology from the nerve and its certificates (theorem 15.3); the Hardy–Weinberg computation (theorem 15.5).

(J) **Hallucination**: a three-token model in which hallucination is a gluing failure with a certificate (proposition 16.1), and the window shadows' Beck–Chevalley defect measures non-Markov coupling (proposition 16.2); its tropical shadow is support-level only (proposition 16.4).

1.4 Status

Statement	Status
operator commutation, rate, lax interchange	proved, finite and general
canonical shadows, band, naturality	proved
fibres, martingale, location of the (U) and (M) defects	proved
lifting functor, quintet double category	proved
a 2-category in which D, C_3 and the laws act monotonically	no-go for coupling liftings (theorem 6.18); open beyond (open problem 6.21)
tropical tower, fibre lifting, tropical secants, Berkovich square	proved
tropical identifiability inside a model	open (open problem 12.5)
wall-crossing formula for Gaussian mixture towers	proposed (conjecture 12.4)
L^2 presheaf: Beck–Chevalley, gluing, comparisons are restrictions	proved
fibre cohomology via assignment-cell nerves; certificates determine it	proved; Hardy–Weinberg computed
lifting obstructions inside a model as cohomology classes	open (open problem 12.7)
three-token hallucination: gluing failure, window Beck–Chevalley, tropical shadow	proved
window defect on real model outputs (pre-registered Study 1)	nonzero in 3 of 5 tasks
obstruction below $\frac{1}{3}$ in pairwise-elicited model answers	kit ready (Study 3); needs model access

1.5 Conventions

X and B denote finite sets, and D is the finite distribution monad with $\eta_X(x) = \delta_x$ and $\mu_X(\Pi) = \sum_\rho \Pi(\rho)\rho$. We identify $\mathsf{D}B$ with the simplex $\Delta_B \subset \mathbb{R}^B$, with the ℓ^1 metric. A *finite probability space* (I, p) has $p(i) > 0$ for all i, and its σ-algebras are partitions. For a partition A, E_A is conditional expectation, acting coordinatewise on $\mathbb{R}^X$-valued functions: $(E_A Z)(i) = \sum_{j \in A(i)} p(j)Z(j)/p(A(i))$, where $A(i)$ is the cell of i. In the tower paper [1], the letters L and R also denote the two sides of the (Cm) diagram. Here L_c and R_c always denote shadows (definition 4.8).

2 The posterior tower

2.1 Levels and contexts

For a finite set B, the iterated distribution spaces $\mathsf{D}B, \mathsf{D}^2B, \mathsf{D}^3B, \ldots$ carry barycentre maps satisfying the monad laws. A hypothesis is $h = (\theta, r) \in \mathsf{Hyp}B = \Theta \times \mathsf{D}B$, with predictive $q(h) = r$. Hyp is a functor and q is natural [1, Def. 3.1]. The shadow space is $\mathsf{Sh}(B) = \mathsf{D}(\mathsf{Hyp}B)$. A context is $c = (\rho, \Pi, \Omega, \Sigma)$ with $\rho \in \mathsf{D}B$, $\Pi \in \mathsf{D}^2B$, $\Omega \in \mathsf{D}^3B$ and $\Sigma \in \mathsf{Sh}B$.

Definition 2.1 (Coherent context)**.** c is *coherent* if

$$\mu(\Pi) = \rho, \qquad \mu(\Omega) = \Pi, \qquad \mathsf{D}(q)(\Sigma) = \Pi. \tag{1}$$

Remark 2.2 (Typing)**.** The third condition is meaningful because q lands in $\mathsf{D}B$, the underlying space of Π. Any variant of the shadow component must come with its own projection to $\mathsf{D}B$ and its action on maps $f\colon B \to B'$. For $\mathsf{Hyp} = \Theta \times \mathsf{D}(-)$ this action is $\mathsf{Hyp}(f)(\theta, r) = (\theta, f_* r)$.

2.2 The absolute tower and the two laws

The absolute tower $\mathsf{C}_3X = X \times \mathsf{D}X \times \mathsf{D}^2X \times \mathsf{D}^3X \times \mathsf{Sh}X$ is a comonad, with comultiplication $\delta(x,c) = ((x,c), \mathsf{E}_3(j_c)(c))$, where $j_c(y) = (y,c)$ [1, §6]. Applying it at $\mathsf{D}X$ gives $\mathsf{C}_3(\mathsf{D}X) = \mathsf{D}X \times \mathsf{D}^2X \times \mathsf{D}^3X \times \mathsf{D}^4X \times \mathsf{Sh}(\mathsf{D}X)$. For a sample $M = \sum_i a_i \delta_{(x_i,c_i)}$:

- the *shift law* λ^σ keeps each sample's context entries, moving them one level up;
- the *averaging law* $\bar{\lambda}$ pushes contexts through the unit and averages them.

2.3 Beck axioms

A mixed distributive law $\lambda\colon \mathsf{DC} \Rightarrow \mathsf{CD}$ must satisfy the unit (U), counit (Co), multiplication (M) and comultiplication (Cm) diagrams [1, Def. 2.1]. The shift law satisfies (Co) and fails (U), (M) and (Cm). The averaging law satisfies (U), (Co) and (M) and fails (Cm) [1, Thms. 8.3, 8.6, 8.7]. The base-relative tower has a strict law [1, Thm. 9.8]. A failure produces two parallel composites with the same source and target, and the comparison between them is the *Beck defect*.

3 Algebra of conditional expectations

3.1 The partition lattice

Partitions of I form a lattice under refinement. The *join* $A \vee B$ has cells $a \cap b$. The *meet* $A \wedge B$ is the finest common coarsening. As σ-algebras, $A \vee B = \sigma(A \cup B)$ and $A \wedge B = A \cap B$.

Lemma 3.1 (The meet is π_0 of a bipartite graph)**.** *Let $G(A,B)$ be the bipartite graph on the cells of A and of B, with an edge a–b whenever $a \cap b \neq \emptyset$. The cells of $A \wedge B$ are the unions over the connected components of $G(A,B)$, and $L^2(A) \cap L^2(B) = L^2(A \wedge B)$.*

Proof. A set lies in $A \cap B$ iff it is a union of A-cells and a union of B-cells, i.e. iff it is closed under the edge relation. A function lies in $L^2(A) \cap L^2(B)$ iff it is constant on cells of both partitions, i.e. iff it is constant on components. □

3.2 The commutation theorem

Write $N_{ab} = p(a \cap b)$ for the contingency matrix and $K = D_A^{-1/2} N D_B^{-1/2}$, with $D_A = \mathrm{diag}(p(a))$ and $D_B = \mathrm{diag}(p(b))$.

Lemma 3.2 (A sufficient pair)**.** *On any probability space, let $C \subseteq A \cap B$ be sub-σ-algebras such that A and B are conditionally independent given C. Then $E_A E_B = E_C = E_B E_A$, and $C = A \cap B$ up to null sets.*

Proof. For $V \in L^2(B)$ and $U \in L^2(A)$, conditional independence gives $\mathbb{E}[UV] = \mathbb{E}[\mathbb{E}[U \mid C]\mathbb{E}[V \mid C]] = \mathbb{E}[U\,\mathbb{E}[V \mid C]]$, using $C \subseteq A$. So $E_A V = E_C V$, and hence $E_A E_B = E_C E_B = E_C$, using $C \subseteq B$. The other order is symmetric. The range of E_C is $L^2(C)$, and that of the commuting product is $L^2(A) \cap L^2(B) = L^2(A \cap B)$. □

Theorem 3.3 (Commutation)**.** *For partitions A, B of a finite probability space the following are equivalent:*

(i) $E_A E_B = E_B E_A$;

(ii) $E_A E_B = E_{A \wedge B}$;

(iii) A *and* B *are conditionally independent given* $A \wedge B$;

(iv) on each component γ *of* $G(A, B)$*, the block* $N|_\gamma$ *has rank one, i.e.* $p(a \cap b)p(\gamma) = p(a)p(b)$ *for all cells* $a, b \subseteq \gamma$;

(v) every singular value of K *is* 0 *or* 1.

The equivalences (i)⇔(ii)⇔(iii) hold on every probability space, with $A \wedge B$ *read as* $A \cap B$*. The value* 1 *in (v) has multiplicity equal to the number of components. Nestedness (*$A \subseteq B$ *or* $B \subseteq A$*) implies (iv).*

Proof. (iii)⇒(ii) is lemma 3.2. (ii)⇒(i): $E_{A \wedge B}$ is self-adjoint, so $E_B E_A = (E_A E_B)^* = E_A E_B$. (i)⇒(ii): the product of commuting orthogonal projections is the projection onto the intersection of their ranges, which is $L^2(A \wedge B)$ by lemma 3.1. (ii)⇒(iii): with $C = A \wedge B$, $\mathbb{E}[UV \mid C] = \mathbb{E}[U\, E_A V \mid C] = \mathbb{E}[U\, E_C V \mid C] = E_C U \cdot E_C V$, since $E_A V = E_A E_B V = E_C V$ is C-measurable.

(iii)⇔(iv): on a finite space it suffices to test indicators. Cells lie inside single components, so conditional independence given C reads $p(a \cap b \mid \gamma) = p(a \mid \gamma)p(b \mid \gamma)$. This says $N|_\gamma$ is the outer product of its margins divided by its total, which for a nonnegative matrix is the same as rank one.

(ii)⇔(v): in the orthonormal bases $1_a/\sqrt{p(a)}$ and $1_b/\sqrt{p(b)}$, $E_A|_{L^2(B)}$ has matrix K, whose singular values are the cosines of the principal angles between $L^2(A)$ and $L^2(B)$. The value 1 occurs exactly on $L^2(A \wedge B)$. Condition (ii) holds iff all other angles are $\pi/2$. If $A \subseteq B$, every A-cell is a component and (iv) is trivial. □

Check A1 confirms the five-way equivalence on 600 random instances, 449 commuting and 151 not.

Proposition 3.4 (The join identity)**.** $E_A E_B = E_{A \vee B}$ *holds iff* $A = B$ *up to null sets.*

Proof. The range of $E_A E_B$ lies in $L^2(A)$. If it equals $L^2(A \vee B) \supseteq L^2(B)$, then $B \subseteq A$, so $E_A E_B = E_B$ and $E_{A \vee B} = E_A$, whence $A = B$. □

Example 3.5 (A non-commuting pair)**.** Let $I = \{1, 2, 3, 4\}$ with $p = (\frac{3}{8}, \frac{1}{8}, \frac{1}{8}, \frac{3}{8})$, $Z(i) = i$, $A_L = \{\{1, 2\}, \{3, 4\}\}$ and $A_R = \{\{1, 3\}, \{2, 4\}\}$. Then

$$(E_{A_R} E_{A_L} Z)(1) = \tfrac{3}{4} \cdot \tfrac{5}{4} + \tfrac{1}{4} \cdot \tfrac{15}{4} = \tfrac{15}{8}, \qquad (E_{A_L} E_{A_R} Z)(1) = \tfrac{3}{4} \cdot \tfrac{3}{2} + \tfrac{1}{4} \cdot \tfrac{7}{2} = 2.$$

$G(A_L, A_R)$ is a 4-cycle, so the meet is trivial. The block $N = \left(\begin{smallmatrix} 3/8 & 1/8 \\ 1/8 & 3/8 \end{smallmatrix}\right)$ has rank two, and K has singular values 1 and $\frac{1}{2}$ (check A3).

3.3 Worked regimes

- *Committed* ($A = B$): the operators coincide. This is a consistency check only.
- *Inherited* ($A \subseteq B$ or $B \subseteq A$): the operators commute with product the coarser one.
- *General*: commutation holds iff (iv) of theorem 3.3 holds. Adding labels does not by itself produce conditional independence. Example 3.5 shows the generic outcome.

3.4 The general case: rate and lax interchange

Theorem 3.6 (Alternating projections, exact rate)**.** *Let c be the cosine of the Friedrichs angle between $L^2(A)$ and $L^2(B)$. In the finite case, c is the largest singular value of K strictly below 1, or 0 if there is none. Then*

$$\|(E_A E_B)^n - E_{A\wedge B}\| = c^{2n-1} \qquad (n \geq 1).$$

On a general probability space the identity holds with the Friedrichs cosine. Then $c < 1$ iff $L^2(A) + L^2(B)$ is closed. Otherwise $(E_A E_B)^n \to E_{A\wedge B}$ only strongly, and the convergence can be arbitrarily slow.

Proof. The identity is the Kayalar–Weinert theorem [30], with $L^2(A) \cap L^2(B) = L^2(A \wedge B)$. Strong convergence is von Neumann's theorem [38], and arbitrarily slow convergence is in [12]. □

Check A4 verifies the rate to 10^{-9} for $n \leq 6$ on 157 non-commuting pairs.

Definition 3.7 (Convex order)**.** For finitely supported laws Q, Q' on a convex set, $Q \leq_{\mathrm{cx}} Q'$ if $\int \varphi\, dQ \leq \int \varphi\, dQ'$ for all convex φ. By Strassen's theorem [37] this holds iff $Q = \mathrm{Law}(\mathbb{E}[U' \mid G])$ for some $U' \sim Q'$ and some sub-σ-algebra G.

Corollary 3.8 (Lax interchange)**.** $E_{A\wedge B} E_A E_B = E_{A\wedge B} = E_{A\wedge B} E_B E_A$. *Hence*

$$\mathrm{Law}(E_{A\wedge B} Z) \leq_{\mathrm{cx}} \mathrm{Law}(E_A E_B Z) \leq_{\mathrm{cx}} \mathrm{Law}(E_B Z) \leq_{\mathrm{cx}} \mathrm{Law}(Z),$$

and symmetrically through $E_B E_A Z$. The laws $\mathrm{Law}((E_A E_B)^n Z)$ decrease in convex order, and

$$W_2\big(\mathrm{Law}((E_A E_B)^n Z), \mathrm{Law}(E_{A\wedge B} Z)\big) \leq c^{2n-1} \|Z - E_{A\wedge B} Z\|_{L^2(p)}.$$

Proof. The tower property gives the first identity. Each displayed law is that of a conditional expectation of the next on the same space. For the rate, $(E_A E_B)^n - E_{A\wedge B} = ((E_A E_B)^n - E_{A\wedge B})(I - E_{A\wedge B})$; apply theorem 3.6 and use the identity coupling to bound W_2. □

So without nestedness or conditional independence, interchange does not hold strictly. It holds through the cospan $E_A E_B \Rightarrow E_{A\wedge B} \Leftarrow E_B E_A$ of convex-order comparisons. These are identities exactly under theorem 3.3, and they shrink at the rate of theorem 3.6.

3.5 Conditioning on a third variable

Proposition 3.9. *Let $X_L \perp X_R \mid H$ on a finite space with full support. Then:*

(a) $E_{\sigma(X_L,H)} E_{\sigma(X_R,H)} = E_{\sigma(H)} = E_{\sigma(X_R,H)} E_{\sigma(X_L,H)}$*;*

(b) $E_{\sigma(X_L)}$ *and* $E_{\sigma(X_R)}$ *commute iff $X_L \perp X_R$ marginally. With full support the meet is trivial, so this fails as soon as both channels depend on H.*

Proof. (a) is lemma 3.2 with $C = \sigma(H)$. (b) With full support, $G(\sigma(X_L), \sigma(X_R))$ is complete bipartite, hence connected; apply theorem 3.3. □

Conditional independence given a third variable is therefore not a commutation hypothesis. The conditioning variable must be built into both σ-algebras (check A7).

4 Shadows

4.1 Label-defined shadows on laws

Let $S = \mathsf{D}B$ with its Borel σ-algebra, and let $a\colon S \to I_a$ be a measurable *label map* with finite codomain. For $\nu \in \mathsf{D}(S)$ with identity variable Z, put $m_a = \mathbb{E}_\nu[Z \mid \sigma(a)]$ and define the label shadow $P_a(\nu) = (m_a)_*\nu$.

Proposition 4.1 (Properties of a single label shadow)**.** *(a)* $\int \rho\, dP_a(\nu) = \int \rho\, d\nu$*, and* $P_a(\nu) \le_{\mathrm{cx}} \nu$*.*

(b) If $a(m_a(\rho)) = a(\rho)$ *for* ν*-a.e.* ρ *(*label preservation*), then* $P_a(P_a\nu) = P_a\nu$*. Without label preservation, the first barycentre can move a point into another label class, and idempotence can fail.*

Proof. (a) is the tower property together with Strassen's theorem. (b) Write $Y = m_a(Z)$. Then Y is $\sigma(a(Z))$-measurable, and label preservation gives $a(Y) = a(Z)$, so $\mathbb{E}[Y \mid a(Y)] = Y$. □

Proposition 4.2 (Composites of label shadows are single poolings)**.** *For label maps* a, b*,*

$$P_b(P_a(\nu)) = \mathrm{Law}\left(E_{A'}Z\right), \qquad A' = \sigma(b \circ m_a) \subseteq \sigma(a).$$

So P_bP_a *pools once, onto a coarsening of the* first *partition. The equation* $P_bP_a = P_aP_b$ *compares coarsenings of two different partitions. It is not the operator identity* $E_AE_B = E_BE_A$*, and theorem 3.3 does not govern it.*

Proof. m_a is constant on $\sigma(a)$-cells, so $b \circ m_a$ is too. Pooling the atoms of $P_a\nu$ along b merges whole cells, and a barycentre of barycentres is the barycentre of the union. □

Example 4.3 (Threshold labels)**.** On $\mathsf{D}\{0,1\} \cong [0,1]$, let ν be uniform on $\{0.1, 0.3, 0.7, 0.9\}$, $a = [s \ge \frac{1}{2}]$ and $b = [s \ge \frac{1}{4}]$. Then $P_b(P_a\nu) = \frac{1}{2}\delta_{0.2} + \frac{1}{2}\delta_{0.8}$, whereas $P_a(P_b\nu) = \frac{1}{4}\delta_{0.1} + \frac{3}{4}\delta_{19/30}$. The fixed-space operators also fail to commute, but with different values (checks C2, C3).

4.2 Labelled random measures

Definition 4.4. A *labelled random measure* over X is a triple (I, p, Z): a finite probability space and a map $Z\colon I \to \mathsf{D}X$. A morphism $f\colon (I,p,Z) \to (I',p',Z')$ is a map with $f_*p = p'$ and $Z' \circ f = Z$. The functor $\mathrm{Law}(I,p,Z) = Z_*p \in \mathsf{D}^2X$ forgets the labels.

Proposition 4.5 (Shadows on a fixed space)**.** *For partitions* A, B *of* I*:*

(a) E_A *is idempotent, preserves* $\mathbb{E}Z$*, and satisfies* $\mathrm{Law}(E_AZ) \le_{\mathrm{cx}} \mathrm{Law}(Z)$*;*

(b) composition of shadows is composition of conditional expectations;

(c) $\mathrm{Law}(E_AE_BZ) = \mathrm{Law}(E_BE_AZ)$ *for all* Z *iff (i)–(v) of theorem 3.3 hold (when* $|X| \ge 2$*).*

Proof. (a) and (b) are immediate. For "only if" in (c): the energy $\int \|v\|^2\, d\,\mathrm{Law}(\cdot)$ is a function of the law. Write $Z = \rho_0 + Y$ with ρ_0 in the relative interior and Y small with zero coordinate sums. Then $\|E_AE_BY\| = \|E_BE_AY\|$ on an open set, hence everywhere, and polarisation gives $S := E_BE_AE_B = E_AE_BE_A$. The range of S lies in $L^2(M)$ with $M = A \wedge B$, and S is self-adjoint, so $S = E_MSE_M = E_M$. For $Y \in L^2(B)$ this reads $\|E_AY\| = \|E_MY\|$, so $E_AY = E_MY$ by Pythagoras, which is (ii). □

Proposition 4.6 (Naturality)**.** *Let $f\colon (I,p)\to(I',p')$ be measure preserving, and let $A=f^{-1}A'$, $B=f^{-1}B'$. Then $E_A(Z'\circ f)=(E_{A'}Z')\circ f$, and likewise for B.*

Proof. $f^{-1}(a')$ is a cell of A of mass $p'(a')$, and the average of $Z'\circ f$ over it is the average of Z' over a' (check G2). □

4.3 The star space of a context and the canonical shadows

Definition 4.7 (Star space)**.** For a coherent context c, let $\kappa(\theta\mid r)=\Sigma(\theta,r)/\Pi(r)$. The *star space* of c has points $(\widetilde{\Pi},r,\theta,x)$ with weights

$$p_c(\widetilde{\Pi},r,\theta,x)=\Omega(\widetilde{\Pi})\,\widetilde{\Pi}(r)\,\kappa(\theta\mid r)\,r(x),$$

restricted to positive weights. Write $\tilde r$ for the coordinate r and $Z=\delta_x$.

Given $\tilde r$, the three *leaves* $\widetilde{\Pi}$, θ and x are independent. The marginals recover the context: $\mathrm{Law}(\widetilde{\Pi})=\Omega$, $\mathrm{Law}(\tilde r)=\Pi$, $\mathrm{Law}(\theta,\tilde r)=\Sigma$ and $\mathrm{Law}(x)=\rho$ (check B1). The context itself does not couple Ω with Σ; the conditional product over $\tilde r$ is the canonical coupling.

Definition 4.8 (Canonical shadows)**.** For a set $T\subseteq\{\mathrm{lev2},\mathrm{hyp},\mathrm{obs}\}$ of leaves, let

$$G_T=\sigma(\tilde r,\ \text{leaves in } T),\qquad S_T=E_{G_T}.$$

The *left shadow* is $L_c=S_{\{\mathrm{lev2}\}}$ and the *right shadow* is $R_c=S_{\{\mathrm{hyp}\}}$.

Theorem 4.9 (The shadow algebra is a commutative band)**.** *$S_TS_U=S_{T\cap U}=S_US_T$ for all leaf sets T,U. In particular $L_cR_c=R_cL_c=E_{\sigma(\tilde r)}$.*

Proof. $G_{T\cap U}\subseteq G_T\cap G_U$. The leaves in $T\setminus U$ and in $U\setminus T$ are independent given $\tilde r$ and the leaves in $T\cap U$, so lemma 3.2 applies. □

Check B2 verifies all 8×8 products on 60 random coherent contexts. The homomorphism $T\mapsto S_T$ need not be injective: if θ is a function of $\tilde r$, then $R_c=S_\emptyset$.

Proposition 4.10 (Dropping the centre)**.** *The label-only shadows $E_{\sigma(\widetilde{\Pi})}$ and $E_{\sigma(\theta)}$ commute iff theorem 3.3 holds for them. They fail to commute in 40 of 60 random contexts (check B3).*

Proposition 4.11 (Markov absorption)**.** *For $Z=\delta_x$: (a) $R_cZ=E_{\sigma(\tilde r)}Z=\tilde r$; (b) $E_{\sigma(\theta)}Z=\sum_r\Sigma(r\mid\theta)\,r$, and in general $E_{\sigma(\tilde r)}E_{\sigma(\theta)}Z\neq E_{\sigma(\theta)}Z$ (59 of 60 random contexts, check B5).*

Proof. (a) holds because $x\perp\theta\mid\tilde r$; (b) is a direct computation. □

The hypothesis level is bookkeeping: given the predictive, it does not move the prediction.

5 Topology and order: how the tower acts

5.1 Two idempotents

The unit laws $\mu\circ\eta_{\mathsf D}=\mathrm{id}=\mu\circ\mathsf D(\eta)$ give one retraction $\mu\colon\mathsf D^2X\to\mathsf DX$ with two sections, and hence two split idempotents on $\mathsf D^2X$:

$$e_\perp=\eta_{\mathsf DX}\circ\mu\ \ (\Pi\mapsto\delta_{\mu\Pi}),\qquad e_\top=\mathsf D(\eta_X)\circ\mu\ \ (\Pi\mapsto\textstyle\sum_x(\mu\Pi)(x)\,\delta_{\delta_x}).$$

Proposition 5.1 (Two idempotents). *(a) $e_\perp$ and $e_\top$ are idempotent, natural in X, and split through the same retraction μ.*

(b) They form a left-zero band: $e_\perp e_\top = e_\perp$ and $e_\top e_\perp = e_\top$.

(c) $e_\top$ is affine; $e_\perp$ is not.

(d) For W_1 built from ℓ^1: μ is 1-Lipschitz, η and $\mathsf{D}(\eta)$ are isometries, and $e_\perp$, $e_\top$ are 1-Lipschitz, hence weakly continuous.

Proof. (a) These are the unit laws and naturality of η and μ. (b) $\eta\mu\mathsf{D}(\eta)\mu = \eta\mu$ and $\mathsf{D}(\eta)\mu\eta\mu = \mathsf{D}(\eta)\mu$. (c) μ and $\mathsf{D}(\eta)$ are affine. η is not, since $\delta_{s\rho+(1-s)\rho'} \neq s\delta_\rho + (1-s)\delta_{\rho'}$. (d) $r \mapsto \langle g, r\rangle$ is 1-Lipschitz for $g \in [-1,1]^X$, so $\|\mu\Pi - \mu\Pi'\|_1 \leq W_1(\Pi, \Pi')$. Also $W_1(\delta_\rho, \delta_{\rho'}) = \|\rho - \rho'\|_1$, and distinct vertices are at distance 2, so $W_1(\mathsf{D}(\eta)\rho, \mathsf{D}(\eta)\rho') = \|\rho - \rho'\|_1$. □

The pooling idempotent of the tower paper is $e = e_\perp$ [1, §13.6], [3].

5.2 Barycentre fibres

Proposition 5.2 (Fibres are convex-order intervals). *For $\rho \in \mathsf{D}X$, let $F_\rho = \mu^{-1}(\rho)$ be the set of Borel probability measures on Δ_X with barycentre ρ. Then:*

(a) F_ρ is convex and weakly compact, and its finitely supported elements are dense in it;

(b) $e_\perp\Pi = \delta_\rho \leq_{\mathrm{cx}} \Pi \leq_{\mathrm{cx}} \mathsf{D}(\eta)\rho = e_\top\Pi$ for every $\Pi \in F_\rho$;

(c) $\mathsf{D}(\eta)\rho$ is the unique element of F_ρ supported on the vertices.

Proof. (a) Prokhorov's theorem and continuity of μ give compactness; F_ρ is convex because μ is affine. For density, partition Δ_X into cells of diameter $< \varepsilon$ and pool Π within each cell. This keeps the barycentre and moves Π by less than ε in W_1. (b) $\rho \mapsto r \sim \Pi$ and $r \mapsto \delta_x$ with probability $r(x)$ are martingale couplings (check B6). (c) Vertices are affinely independent. This is the Choquet–Meyer uniqueness of maximal measures on a simplex [36]. □

5.3 A coherent context is a martingale

Theorem 5.3 (The tower as a martingale). *On the star space let $\mathcal{F}_0$ be trivial, $\mathcal{F}_1 = \sigma(\widetilde{\Pi})$, $\mathcal{F}_2 = \sigma(\widetilde{\Pi}, \tilde{r})$ and $\mathcal{F}_3 = \sigma(\widetilde{\Pi}, \tilde{r}, x)$, and put $M_k = \mathbb{E}[Z \mid \mathcal{F}_k]$. Then (M_k) is a Δ_X-valued martingale with*

$$\mathrm{Law}(M_0) = \delta_\rho, \quad \mathrm{Law}(M_1) = \mathsf{D}(\mu)\Omega, \quad \mathrm{Law}(M_2) = \Pi, \quad \mathrm{Law}(M_3) = \mathsf{D}(\eta)\rho,$$

a chain $\delta_\rho \leq_{\mathrm{cx}} \mathsf{D}(\mu)\Omega \leq_{\mathrm{cx}} \Pi \leq_{\mathrm{cx}} \mathsf{D}(\eta)\rho$ in F_ρ. The hypothesis branch does not move it: $\mathbb{E}[Z \mid \theta, \tilde{r}] = M_2$.

Proof. $M_0 = \rho$, $M_1 = \mu(\widetilde{\Pi})$, $M_2 = \tilde{r}$ and $M_3 = \delta_x$. The last statement is proposition 4.11 (check B4). □

So the levels of the tower are the successive conditional expectations of the observation along one filtration. Coherence is the martingale property in law. Nested shadows land on levels and commute.

5.4 The Beck defects located

Proposition 5.4 (Where the defects of λ^σ sit)**.** *At level one:*

(a) the two sides of (U) at (x, c) are $e_\perp \Pi = \delta_\rho$ and $e_\top \Pi = \mathsf{D}(\eta)\rho$, so the unit defect is the full convex-order diameter of F_ρ;

(b) for $M' = \sum_\ell \nu_\ell \delta_{M_\ell}$ with $M_\ell = \sum_k a_{\ell k} \delta_{(x_{\ell k}, c_{\ell k})}$, on the sample-index space $I = \{(\ell, k)\}$ with $p = \nu_\ell a_{\ell k}$ and $W(\ell, k) = \rho_{\ell k}$, the two sides of (M) are $\mathrm{Law}(W)$ *(λ-first) and* $\mathrm{Law}(\mathbb{E}[W \mid \ell])$.

Proof. This is [1, Prop. 13.8] read in the present notation, and it is verified with the tower code's own λ^σ, C_3, η and μ (check B7). □

The shift law pools after the unit and spreads after the multiplication. Every order that reads conditional expectation monotonically therefore orients the two defects oppositely [1, Prop. 13.9, Thm. 13.15, Prop. 13.17].

5.5 Continuity

Proposition 5.5. *(a) On a fixed index with fixed partitions, $(p, Z) \mapsto \mathrm{Law}(E_A Z)$ is continuous for full-support p, and $W_1(\mathrm{Law}(E_A Z), \mathrm{Law}(E_A Z')) \le \|Z - Z'\|_{L^1(p)}$. (b) Nearest-atom shadows are Borel but not continuous. With atoms $(\frac{2}{3}, \frac{1}{3})$ and $(\frac{1}{3}, \frac{2}{3})$, two inputs at W_1-distance $\frac{1}{1000}$ have outputs at distance $\frac{501}{2000}$ (check C4).*

Proof. (a) Cell averages are continuous where the cell masses are positive, and conditional expectation is an L^1 contraction. (b) A point crossing the Voronoi wall changes cells. □

6 Categorical structure

6.1 Why a separate construction is needed

The operator results act on measures and random variables; context maps act on contexts. A double category needs objects, horizontal and vertical arrows with associative compositions, squares with typed boundaries, identity squares, two compositions of squares, and the interchange law for composable grids. Expressions such as $g \circ L$ or $R \circ f$ are undefined unless a functor connects the two layers. This section builds that functor and then the double category.

6.2 The context category

Definition 6.1 (Context category)**. Ctx** has as objects the coherent contexts c on finite bases B_c. A morphism $f : c \to c'$ is a map $f : B_c \to B_{c'}$ with

$$c' = \big(f_*\rho,\ \mathsf{D}^2 f(\Pi),\ \mathsf{D}^3 f(\Omega),\ \mathsf{Sh} f(\Sigma)\big).$$

Composition and identities are those of maps of finite sets.

Proposition 6.2. Ctx *is a category, and pushing forward along f preserves coherence.*

Proof. D^k and Sh are functors. Coherence is preserved by naturality of μ ($\mu \circ \mathsf{D}^2 f = \mathsf{D} f \circ \mu$) and of q. □

6.3 The lifting functor, in two versions

Remark 6.3 (Two readings of the level-two component)**.** A morphism $f\colon c \to c'$ can carry either of two kinds of level-two data:

(i) a specified map $f_\Pi\colon \mathsf{D}^2(B_c) \to \mathsf{D}^2(B_{c'})$, with $(g \circ f)_\Pi = g_\Pi \circ f_\Pi$ imposed as an axiom (*component-data version*);

(ii) a base map $f\colon B_c \to B_{c'}$, with induced action $\mathsf{D}^2(f)$ (*base-map version*, as in definition 6.1).

The two agree when $f_\Pi = \mathsf{D}^2(f)$. This is a coherence condition, not automatic.

Definition 6.4 (Lifting functor)**.** Define $F(c) = \mathsf{D}^2(B_c)$, with $F(f) = f_\Pi$ in the component-data version and $F(f) = \mathsf{D}^2(f)$ in the base-map version.

Proposition 6.5 (Functoriality, component-data version)**.** *If* $(\mathrm{id}_c)_\Pi = \mathrm{id}$ *and* $(g \circ f)_\Pi = g_\Pi \circ f_\Pi$*, then* F *is a functor.*

Proof. $F(\mathrm{id}_c) = (\mathrm{id}_c)_\Pi = \mathrm{id}_{F(c)}$ and $F(g \circ f) = (g \circ f)_\Pi = g_\Pi \circ f_\Pi = F(g) \circ F(f)$, and all three maps have type $\mathsf{D}^2(B_c) \to \mathsf{D}^2(B_{c''})$. □

Proposition 6.6 (Functoriality, base-map version)**.** *In the base-map version* F *is a functor.*

Proof. D^2 is a functor. □

Remark 6.7. In the component-data version f_Π is already a map between level-two spaces. Applying D^2 to it again would give a map $\mathsf{D}^4(B_c) \to \mathsf{D}^4(B_{c'})$ of the wrong type, so $F(f) = f_\Pi$ is the type-correct choice there.

6.4 Shadows along context maps

The canonical shadows live on star spaces, not on $F(c)$. The star construction is functorial on a large subcategory.

Proposition 6.8 (Star spaces are functorial)**.** *Let* $f\colon c \to c'$ *be a morphism of* **Ctx** *such that* $\mathsf{D}f$ *is injective on* $\operatorname{supp}\Pi$*. Then*

$$\tilde{f}(\widetilde{\Pi}, r, \theta, x) = \big(\mathsf{D}^2 f(\widetilde{\Pi}),\ \mathsf{D}f(r),\ \theta,\ f(x)\big)$$

maps the star space of c *measure-preservingly onto the star space of* c'*. It pulls back* G_T *to* G_T *for* $T \subseteq \{\mathrm{lev2}, \mathrm{hyp}\}$*. Hence* L_c *and* R_c *are natural:* $L_c(Z' \circ \tilde{f}) = (L_{c'} Z') \circ \tilde{f}$*, and likewise for* R*.*

Proof. Summing p_c over the fibre of $\tilde{f}$ gives $\Omega(\widetilde{\Pi})\widetilde{\Pi}(r)\kappa(\theta \mid r) \sum_{x \in f^{-1}(x')} r(x)$. By injectivity on atoms, each factor equals its primed counterpart. Injectivity also gives $\tilde{f}^{-1}\sigma(\mathsf{D}f(\tilde{r})) = \sigma(\tilde{r})$ and $\tilde{f}^{-1}\sigma(\mathsf{D}^2 f(\widetilde{\Pi})) = \sigma(\widetilde{\Pi})$, while θ is unchanged. Naturality is proposition 4.6 (check G3, on 36 context maps). For $T \ni \mathrm{obs}$ only $\tilde{f}^{-1} G'_T \subseteq G_T$ holds, because f may merge outcomes. □

So the "(L, R)-compatible subcategory" of a context-map formalism is not hypothetical. It contains every context map that is injective on predictive atoms: relabellings of latent classes, coarsenings of the observation space that keep the atoms distinct, and their composites.

6.5 Typed squares and quintets

Definition 6.9 (Typed square data)**.** A square with boundary ($h\colon a \to b$, $v\colon a \to c$, $w\colon b \to d$, $k\colon c \to d$) may carry a comparison α only between the composites $k \circ v$ and $w \circ h$, which share source and target. The boundary, the comparison, and the composition laws are all part of the data.

Proposition 6.10 (Quintets of a locally preordered 2-category)**.** *Let $\mathcal{K}$ be a 2-category whose hom-categories are preorders. The quintet double category $Q(\mathcal{K})$ has the objects and 1-cells of $\mathcal{K}$ in both directions, and exactly one square with boundary (h, v, w, k) iff $k \circ v \leq w \circ h$. Then $Q(\mathcal{K})$ is a strict double category, and interchange, associativity and the unit laws hold automatically.*

Proof. This is Ehresmann's quintet construction [26, 28]. Pasting is transitivity together with monotone whiskering, and uniqueness of 2-cells makes every equation trivial. □

In a locally discrete 2-category such as **Meas**, where the only 2-cells are identities [1, §§13.1, 13.5], squares are commuting squares. Nontrivial cells require an order.

6.6 What the tower supplies, and what it lacks

(a) *Convex order on one level.* $\leq_{\mathrm{cx}}$ is a partial order on $\mathsf{D}(K)$ for convex K, and pushforward along affine maps is monotone. $T \mapsto \mathrm{Law}(S_T Z)$ is a monotone map from the Boolean lattice of leaf sets into $(F_\rho, \leq_{\mathrm{cx}})$. The martingale of theorem 5.3 is a chain of vertical arrows.

(b) *Strict interchange for canonical shadows* (theorem 4.9).

(c) *Lax interchange in general*: a cospan of squares into the meet (corollary 3.8).

(d) *The Beck defects.* The (U) and (M) defects of λ^σ are vertical arrows of opposite orientation, so λ^σ is neither lax nor oplax in any $Q(\mathcal{K})$ whose order reads conditional expectation monotonically. The (Cm) defect of $\bar{\lambda}$ is incomparable in the joint convex order [1, Prop. 13.10]. It is a cell of one orientation for label pooling at every level [1, Thm. 13.16].

(e) *What is missing.* D must act monotonically on hom-preorders. The pointwise convex order fails this, because $\delta_a \leq_{\mathrm{cx}} \delta_b$ forces $a = b$. The candidate is the relation (Barr) lifting [17] of a relation along D. The parts of the structure that are monotone for any relation are settled by lemma 6.11.

For a relation $R \subseteq X \times Y$ between finite sets, its *Barr lifting* $\bar{\mathsf{D}}R \subseteq \mathsf{D}X \times \mathsf{D}Y$ is

$$\Pi\ \bar{\mathsf{D}}R\ \Pi' \iff \text{there is a coupling } \omega \in \mathsf{D}(X \times Y) \text{ of } \Pi \text{ and } \Pi' \text{ with } \operatorname{supp}\omega \subseteq R.$$

Lemma 6.11 (D, η and μ are monotone for the Barr lifting)**.** *Let $R \subseteq X \times Y$ and $S \subseteq Y \times Z$ be relations between finite sets.*

(a) If $R \subseteq R'$ then $\bar{\mathsf{D}}R \subseteq \bar{\mathsf{D}}R'$. Moreover $\bar{\mathsf{D}}(\Delta_X) = \Delta_{\mathsf{D}X}$ and $\bar{\mathsf{D}}R\,;\bar{\mathsf{D}}S \subseteq \bar{\mathsf{D}}(R\,;S)$. Hence if R is a preorder, so is $\bar{\mathsf{D}}R$.

(b) D is monotone. *If $f\colon X \to X'$ and $g\colon Y \to Y'$ satisfy $(f \times g)(R) \subseteq R'$, then $(\mathsf{D}f \times \mathsf{D}g)(\bar{\mathsf{D}}R) \subseteq \bar{\mathsf{D}}R'$.*

(c) η is monotone and reflects. $\delta_x\ \bar{\mathsf{D}}R\ \delta_y \iff x\,R\,y$.

(d) μ is monotone. *If* $\mathbf{\Pi}\ \bar{\mathsf{D}}(\bar{\mathsf{D}}R)\ \mathbf{\Pi}'$ *in* $\mathsf{D}^2X \times \mathsf{D}^2Y$, *then* $\mu\mathbf{\Pi}\ \bar{\mathsf{D}}R\ \mu\mathbf{\Pi}'$.

Proof. (a) Monotonicity in R is immediate. The only coupling of Π with itself supported on the diagonal is the diagonal coupling, so $\bar{\mathsf{D}}(\Delta_X) = \Delta_{\mathsf{D}X}$. For composition, let ω_1 couple Π, Π' inside R and ω_2 couple Π', Π'' inside S. Glue them along their common marginal Π': $\omega(x,y,z) = \omega_1(x,y)\,\omega_2(y,z)/\Pi'(y)$ for $\Pi'(y) > 0$. The (x,z)-marginal of ω couples Π, Π'', and each point of its support has a witness y with $x\,R\,y\,S\,z$. Reflexivity and transitivity of $\bar{\mathsf{D}}R$ follow. (b) Push a witnessing coupling forward along $f \times g$. (c) The only coupling of two point masses is $\delta_{(x,y)}$. (d) Let Ω couple $\mathbf{\Pi}, \mathbf{\Pi}'$ with support in $\bar{\mathsf{D}}R$. For each $(p,p') \in \operatorname{supp}\Omega$ choose a coupling $\omega_{p,p'}$ of p, p' with support in R. Then $\omega = \sum_{(p,p')} \Omega(p,p')\,\omega_{p,p'}$ has marginals $\sum \Omega(p,p')\,p = \mu\mathbf{\Pi}$ and $\mu\mathbf{\Pi}'$, and its support lies in R. Checks R1–R4 test (a)–(d) on random finite instances and confirm the counterexample below. □

Only the forward direction of (d) is needed for monotonicity; the converse fails in general, because a coupling of the flattened laws need not come from a coupling of the laws on laws. For example, with $R = \Delta_{\{0,1\}}$, the laws $\mathbf{\Pi} = \frac{1}{2}\delta_{\delta_0} + \frac{1}{2}\delta_{\delta_1}$ and $\mathbf{\Pi}' = \delta_{(\delta_0+\delta_1)/2}$ have the same flattening, but $\bar{\mathsf{D}}(\bar{\mathsf{D}}\Delta)$ is equality on D^2X, so they are not related.

6.7 Which lifting? Three facts that sharpen the problem

Write $\hat{c} = \mathsf{E}_3(\eta)(c)$ for a context pushed through the unit, and, for a sample $M = \sum_k a_k\delta_{(x_k,c_k)}$, write $\bar{e} = \sum_k a_k\hat{c}_k$ for the pooled context that $\bar{\lambda}$ produces. The set of contexts is convex (the coherence conditions are linear) and $\mathsf{E}_3(\eta)$ is affine, so the set S of all $\hat{c}$ is convex.

Lemma 6.12 (Coupling liftings)**.** *Call a lifting a* coupling lifting *if* $\Pi \le \Pi'$ *means that some coupling of* Π *and* Π' *is admissible, where the admissible couplings are closed under mixtures and under gluing along a common marginal. Then the conclusions of lemma 6.11(a), (b) and (d) hold: the lifted relation is a preorder, and* D *and* μ *are monotone. Strassen's martingale lifting (admissible means* $\mathbb{E}[\mathit{target} \mid \mathit{source}] = \mathit{source}$*) is a coupling lifting, and on a convex set it is the convex order.*

Proof. The proofs of lemma 6.11(a), (b), (d) use only these two closure properties. A mixture of martingale couplings is a martingale coupling, and so is the composite of two (tower property). Check TC4. □

Lemma 6.13 (The convex order is not a Barr lifting)**.** *No relation* R *on a convex set* K *with at least two points has* $\bar{\mathsf{D}}R = \le_{\mathrm{cx}}$ *on* $\mathsf{D}(K)$.

Proof. A Barr lifting has $\delta_r\ \bar{\mathsf{D}}R\ \delta_{r'}$ iff $r\,R\,r'$ (lemma 6.11(c)), while $\delta_r \le_{\mathrm{cx}} \delta_{r'}$ iff $r = r'$. So R would be equality, whose lifting is equality, but $\delta_{(r+r')/2} \le_{\mathrm{cx}} \frac{1}{2}\delta_r + \frac{1}{2}\delta_{r'}$. Check TC3. □

Proposition 6.14 (The comultiplication cell of $\bar{\lambda}$)**.** *At level one, the two sides of the (Cm) cell of* $\bar{\lambda}$ *at a sample* M *are laws on pairs (label, context):*

(a) the left side $\delta(\bar{\lambda}M)$ *pairs every label with the pooled context* $\bar{e}$*; the right side* $\mathsf{C}_3(\bar{\lambda})(\bar{\lambda}(\mathsf{D}\delta\,M))$ *pairs each label coming from item* k *with that item's own context* $\hat{c}_k$*; the two label marginals agree (check TC1).*

Consider liftings that preserve labels, as cells over $\mathsf{C}_3(\varepsilon)$ *must, since the two sides agree after* $\mathsf{C}_3(\varepsilon)$ *[1, Thm. 8.7]. Then a comparison between the two sides must use the coupling that sends each* $(\mathit{label}, \hat{c}_k)$ *to* $(\mathit{label}, \bar{e})$*, and:*

(b) Barr liftings collapse. *If the Barr lifting of a label-preserving relation R orients the cell the same way for every M, then for each label R relates any two points of the relative interior of S in both directions. So the order identifies all interior contexts.*

(c) The martingale lifting orients the cell only without label–context correlation. *The cell is oriented (pooled below unpooled) iff, for every label r, the mean of $\hat{c}_k$ over the items that produce r equals $\bar{e}$. This fails on the coherent sample of example 6.15, holds when every label has the same conditional mean context, and holds in only 30 of 200 random samples (check TC5).*

(d) Forgetting labels, the cell is always oriented*: $\delta_{\bar{e}} \leq_{\mathrm{cx}} \sum_k a_k \delta_{\hat{c}_k}$ for the context marginals.*

Proof. (a) is computed with the tower code. Since the left side carries the single context $\bar{e}$ for each label and the label marginals agree, a label-preserving coupling is forced. (b) Orienting the cell for every M puts $(\hat{c}_k, \sum_l a_l \hat{c}_l)$ in R (or its reverse) for all finite families in S and all weights; that is, $(u, au + (1-a)w) \in R$ for all $u, w \in S$ and $a \in (0,1)$. If u, v lie in the relative interior of S, then $w = (v - au)/(1-a) \in S$ for small $a > 0$, so $u \, R \, v$, and likewise $v \, R \, u$ (check TC2). For the reverse orientation the same argument applies with the pairs reversed. (c) The forced coupling, read from pooled to unpooled, is a martingale coupling iff the conditional mean of the unpooled context given the label is $\bar{e}$. (d) $\sum_k a_k \hat{c}_k = \bar{e}$. □

Example 6.15 (An explicit coherent sample)**.** For $i = 0, 1$ let $c_i = (\delta_i,\ \delta_{\delta_i},\ \delta_{\delta_{\delta_i}},\ \delta_{(t,\delta_i)})$. Each is coherent: $\mu(\delta_{\delta_i}) = \delta_i$, $\mu(\delta_{\delta_{\delta_i}}) = \delta_{\delta_i}$, and the shadow projects to Π. Take $M = \frac{1}{2}\delta_{(0,c_0)} + \frac{1}{2}\delta_{(0,c_1)}$. On the right side of the cell, the label δ_0 comes only from item 0, with context $\hat{c}_0$, and the label δ_1 only from item 1, with context $\hat{c}_1$. The pooled context is $\bar{e} = \frac{1}{2}\hat{c}_0 + \frac{1}{2}\hat{c}_1$, and $\hat{c}_0 \neq \hat{c}_1$ (their first components are δ_{δ_0} and δ_{δ_1}). So the conditional mean context given the label δ_0 is $\hat{c}_0 \neq \bar{e}$, and the martingale criterion of proposition 6.14(c) fails. The two sides of the cell differ, as in [1, Thm. 8.7], whose counterexample has the same label structure but uses contexts that are not coherent. Check TC12 verifies every step in exact arithmetic.

So the order in which $\bar{\lambda}$'s comultiplication cell is oriented exists, but it is the convex order of the context marginal, which forgets how labels and contexts are coupled. This is the precise sense of [1, Prop. 13.10, Thm. 13.16]: incomparable in the joint convex order, oriented for label pooling.

6.8 The comultiplication and a no-go theorem

Monotonicity of the comultiplication $\delta(x, c) = ((x, c), \mathsf{E}_3(j_c)(c))$ needs an order on the inserted labels $j_c(y) = (y, c)$, because the whole context is written into every label. The two natural choices behave oppositely.

Proposition 6.16 (Label-preserving orders make δ monotone)**.** *Order $\mathsf{C}_3 Y$ by $(y, c) \leq (y', c')$ iff $y = y'$ and $c \preccurlyeq c'$, where $\preccurlyeq$ is a preorder on contexts obtained by lifting along E_3 with a coupling lifting on each component (lemma 6.12) that is Barr-type on label coordinates. Then δ is monotone.*

Proof. Let $c \preccurlyeq c'$. The first components $(x, c) \leq (x, c')$ by definition. For the second, $j_c(y) \leq j_{c'}(y)$ for every y, so pushing forward along the pointwise-related maps j_c and $j_{c'}$ gives related laws, by the coupling $(j_c, j_{c'})_*$; hence $\mathsf{E}_3(j_c)(c) \preccurlyeq \mathsf{E}_3(j_{c'})(c)$. Next, $\mathsf{E}_3(j_{c'})$ is monotone, because pushforward along a monotone map preserves a coupling lifting (lemma 6.11(b), lemma 6.12); hence $\mathsf{E}_3(j_{c'})(c) \preccurlyeq \mathsf{E}_3(j_{c'})(c')$. Transitivity, which holds because admissible couplings glue, gives $\mathsf{E}_3(j_c)(c) \preccurlyeq \mathsf{E}_3(j_{c'})(c')$. Check TC8 verifies the first step. □

For instance, two coherent contexts that differ only in their shadows, with Σ_1 a mean-preserving spread of Σ_0 in the descriptor θ, satisfy $c_0 \preccurlyeq c_1$ for the lifting that is a martingale in θ and Barr-type in the label, and their images under δ are related level by level.

Proposition 6.17 (The context-marginal order is not compatible)**.** *Order level-one laws on $\mathsf{C}_3 Y$ by the convex order of their context marginals, the order that orients the comultiplication cell of $\bar{\lambda}$ (proposition 6.14(d)). Then:*

(a) $\delta(x, c_0) \leq \delta(x, c_1)$ at level one only if $c_0 = c_1$, because the level-one output of $\delta(x, c)$ has context marginal δ_c (check TC6) and point masses are comparable in the convex order only when equal;

(b) $\mathsf{D}(\varepsilon)$ is not monotone: two laws with the same context marginal are equivalent, but their label marginals can be incomparable (check TC7).

Call a preorder on contexts *non-degenerate* if no two distinct points of the relative interior of S are equivalent under it, that is, if it is antisymmetric on the relative interior of S. Equality is non-degenerate; an order that makes all interior contexts equivalent is not.

Theorem 6.18 (No-go for coupling liftings)**.** *Suppose the level-one laws on $\mathsf{C}_3 Y$ are ordered by a coupling lifting whose admissible couplings preserve labels and act on the context coordinate either by support in a preorder (Barr type) or by the martingale condition (convex type). If the lifting orients the comultiplication cell of $\bar{\lambda}$ the same way for every sample, then:*

(a) in the Barr case, the context preorder makes any two points of the relative interior of S equivalent, so it is degenerate;

(b) the martingale case is impossible: example 6.15 is a coherent sample at which the cell is not oriented.

If labels are not preserved, monotonicity of $\mathsf{D}(\varepsilon)$ fails whenever labels are discretely ordered, as for the only order that orients the cell for every sample (proposition 6.17(b)). Hence no coupling lifting of these types whose context order is non-degenerate makes D, η, μ, ε and δ monotone and orients the label-pooling cells of $\bar{\lambda}$.

Proof. (a) is proposition 6.14(b): the order contains $(u, au+(1-a)w)$ for all $u, w \in S$ and $a \in (0, 1)$, and any two relative-interior points are connected this way in both directions. This is an interior collapse; the theorem claims nothing about the boundary of S. (b) is proposition 6.14(c) applied to example 6.15. For the last paragraph: a coupling that moves mass between distinct, discretely ordered labels relates laws whose images under $\mathsf{D}(\varepsilon)$ are unrelated, so monotonicity of $\mathsf{D}(\varepsilon)$ forces label preservation; proposition 6.17(b) exhibits the failure for the context-marginal order, and proposition 6.17(a) shows that this order would in addition keep δ monotone only for the equality order on contexts. □

So δ is not the obstruction: label-preserving liftings make it monotone (proposition 6.16). The obstruction is the comultiplication cell itself, which is oriented only when labels are pooled away, and pooling labels away is incompatible with the counit and with δ.

6.9 Escaping thinness: the coupling 2-category

Theorem 6.18 uses thinness: between two 1-cells there is at most one 2-cell, so "oriented" means "some admissible coupling exists". The no-go is escaped by letting 2-cells carry data.

Definition 6.19 (The coupling 2-category)**.** $\mathcal{K}_{\mathrm{coup}}$ has finite sets as objects and Kleisli maps $f\colon A \to \mathsf{D}B$ as 1-cells. A 2-cell $\alpha\colon f \Rightarrow g$ is a family $(\omega_a)_{a\in A}$, where ω_a is a coupling of $f(a)$ and $g(a)$. Vertical composition glues ω_a and ω'_a by the conditional product along their common marginal $g(a)$; the identity on f is the diagonal coupling of $f(a)$ with itself. Postwhiskering by $h\colon B \to C$ pushes each ω_a forward along $h \times h$.

Proposition 6.20 (What holds and what fails)**.** *(a) Vertical composition is associative and unital: it is composition of the Markov kernels that the couplings define.*

(b) Postwhiskering preserves a vertical composite, $h_*(\beta \circ \alpha) = h_*\beta \circ h_*\alpha$, whenever the pushed-forward glued law of (hX, hY, hZ) satisfies $hX \perp hZ \mid hY$, that is, Beck–Chevalley for the windows $\sigma(hX, hY)$ and $\sigma(hY, hZ)$ ($c = 0$). When the two sides differ, that defect is positive; for a non-injective h they differ generically.

(c) In $\mathcal{K}_{\mathrm{coup}}$ the comultiplication cell of $\bar{\lambda}$ is a 2-cell for every sample: the forced coupling of proposition 6.14, sending $(\mathit{label}, \hat{c}_k)$ to $(\mathit{label}, \bar{e})$. Its image under the label projection is the identity 2-cell, as the counit requires.

Proof. (a) A coupling ω of p and q is the same as p with a kernel $p \to q$; the conditional product composes the kernels. The diagonal coupling is the identity kernel. (b) The left side is the (hX, hZ)-marginal of the pushed-forward glued law Q; the right side is the conditional product of its (hX, hY) and (hY, hZ) marginals. Under $hX \perp hZ \mid hY$ these coincide (proposition 16.3(a)); if they differ, the independence fails, so $c > 0$ by theorem 14.4. The original glued law makes $X \perp Z$ given the fine middle Y, which does not survive merging the values of Y. (c) The forced coupling has the two sides as marginals and preserves labels. Checks TC9 (300 cases), TC10 (the two sides differ in all 500 random cases, always with $c > 0$, and agree for laws built conditionally independent given hY) and TC11 (200 samples). □

So in $\mathcal{K}_{\mathrm{coup}}$ orientation is automatic, and the obstruction moves: postwhiskering preserves composites only up to a comparison, and the comparison is measured by the Beck–Chevalley defect of section 14. Treating those comparisons as part of the structure, with coherence laws of their own, needs a third level of cells (a tricategory, or a double category with lax interchange). That is a larger undertaking; the finite questions below come first.

Open problem 6.21 (Beyond coupling liftings)**.** Theorem 6.18 answers open problem 4 of [1] negatively for coupling liftings of Barr and martingale type. Is there an order of another kind (not defined by admissible couplings, for example one that compares joint laws through a family of test functions) that is non-degenerate in the sense defined before theorem 6.18, makes D, η, μ, ε and δ monotone, and orients every label-pooling cell of $\bar{\lambda}$? A positive answer makes $\bar{\lambda}$ a lax mixed distributive law in $Q(\mathcal{K})$. The theorem shows that any such order must compare label and context jointly without being determined by couplings. In the coupling 2-category of section 6.9 the cell always exists; there the question becomes whether the 2-cells of $\bar{\lambda}$ satisfy the remaining coherence axioms (coassociativity with δ) up to the whiskering comparisons of proposition 6.20(b). It does not concern λ^{σ}, which item (d) of section 6.6 excludes in any such order.

7 Beck defects as comparison cells

7.1 What a factorisation requires

Proposition 7.1 (Required data)**.** *To factor a Beck defect through a shadow, one needs:*

(i) *an action of the shadow on the relevant tower objects;*

(ii) *a comparison map through that action;*

(iii) *a typed coherence map completing the factorisation;*

(iv) *a proof that the composite comparison equals the original defect.*

The observation that one side looks pooled and the other unpooled motivates a factorisation but does not prove one. For the defects computed so far, all four items are now supplied.

Proposition 7.2 (Each computed defect is one conditional-expectation step)**.** *On a fixed labelled space built from the data of the axiom:*

(a) *(U) for λ^σ at level one: the space is the star space, $Z = \delta_x$, and the step is E_{triv}. The sides are $\mathrm{Law}(E_{\mathrm{triv}} Z) = \delta_\rho$ and $\mathrm{Law}(Z) = \mathsf{D}(\eta)\rho$.*

(b) *(M) for λ^σ at level one: the space is the sample index, and the step is $E_{\sigma(\ell)}$ (proposition 5.4(b)).*

(c) *(Cm) for $\bar{\lambda}$ at every level and in the shadow: the space is the sample index, and the step is the coordinatewise conditional expectation of the context labels onto the trivial σ-algebra, with the sample coordinate kept [1, Thm. 13.16].*

In each case items (i)–(iv) of proposition 7.1 hold: the shadow acts on the labelled space, the comparison is the martingale coupling, the coherence map is the identity on the index, and the equality is the cited computation.

Remark 7.3 (The two-shadow conjecture)**.** An earlier formulation conjectured that the (Cm) defect factors through a *pair* of shadows. In every computed case one conditional-expectation step suffices. Whether some Beck defect of a tower variant needs two non-commuting shadows, and so carries a genuinely lax interchange (corollary 3.8), is open.

7.2 One-shadow and two-shadow options

Option A keeps one shadow component, $\mathsf{C}_A(Y) = Y \times \mathsf{D}B \times \mathsf{D}^2 B \times \mathsf{D}^3 B \times \mathsf{Sh}(Y)$. Option B keeps two, $\mathsf{C}_B(Y) = \mathsf{C}_A(Y) \times \mathsf{Sh}(\mathsf{D}Y)$.

Proposition 7.4. *Let $\mathsf{C}_E Y = Y \times \prod_i E_i Y$ be built from functors E_i, and let λ be assembled from transports $\lambda_i \colon \mathsf{D}E_i \Rightarrow E_i \mathsf{D}$. Then (U), (Co) and (M) hold for λ iff they hold componentwise. (Cm) does not decompose this way, because $\delta(y, c) = ((y, c), \prod_i E_i(j_c)(c_i))$ feeds the* whole *context into each component.*

Proof. The structure maps of D act factorwise, and equality in a product is componentwise. The formula for δ is [1, §6]. □

So neither option repairs a failed axiom automatically. For (Cm) the averaging-type transport has its defect in every component at once [1, Thm. 13.16].

7.3 Higher cells

Proposition 7.5 (Triviality after equality)**.** *If two parallel morphisms are equal, the identity comparison on them is trivial. In a quintet double category of a locally preordered 2-category, any two parallel 2-cells are equal, so no nontrivial 3-cells arise.*

A nontrivial 3-cell would require hom-categories that are not preorders. It would then be a comparison between two distinct comparisons with the same boundary. No such structure is used or needed here.

8 The measure-theoretic (Giry) extension

Let X be Polish and let $\mathcal{P}(X)$ carry the weak topology. Then $\mathcal{P}$ is a monad on Polish spaces, with $\eta_X(x) = \delta_x$ and $\mu_X(\Pi)(A) = \int \rho(A)\, d\Pi(\rho)$, both continuous [27]. On finite discrete X it is D. A category of closed convex subsets of Euclidean spaces is not closed under $\mathcal{P}$, so the continuous tower must use spaces of measures as objects. A measure-valued shadow needs five ingredients, and each is now available:

(i) *a measurable space of measures*: $\mathcal{P}(X)$ with its Borel structure, which is standard Borel when X is;

(ii) *a conditional barycentre*: for $Z\colon \Omega \to \mathcal{P}(X)$ and $G \subseteq \mathcal{F}$, set $\mathbb{E}[Z \mid G](A) = \mathbb{E}[Z(A) \mid G]$; when X is standard Borel this has a version that is a random probability measure;

(iii) *measurability*: the version can be chosen measurable;

(iv) *integrability*: automatic, since values are probability measures;

(v) *structure*: propositions 4.5 and 4.6 hold verbatim on a fixed probability space.

The commutation results extend as follows.

- (i)⇔(ii)⇔(iii) of theorem 3.3 hold for sub-σ-algebras of any probability space, applied to $Z(A)$ for A in a countable generating π-system.
- The finite criteria (iv) and (v) are replaced by the spectral condition $E_A E_B E_A = E_{A\wedge B}$ on $L^2(A)$. The meet need not be countably generated.
- Theorem 3.6 holds with c the Friedrichs cosine. If $c = 1$, the lax interchange holds only in the strong limit.
- For compact metric X, proposition 5.2(a,b) and theorem 5.3 hold with $\mathcal{P}(\eta)\rho$ as top. Since $\mathcal{P}(X)$ is a Bauer simplex, maximal representing measures are unique [36].

Open problem 8.1 (Lax interchange without a rate)**.** When $c = 1$, does the cospan of corollary 3.8 admit a coherent choice, natural along measure-preserving maps that do not pull back the σ-algebras?

9 The finite stochastic-to-convex functor

Definition 9.1. FinStoch has as objects finite sets and as morphisms stochastic kernels $K\colon X \to Y$, i.e. $K(y \mid x) \geq 0$ with $\sum_y K(y \mid x) = 1$. Composition is $(L \circ K)(z \mid x) = \sum_y L(z \mid y)K(y \mid x)$, and the identities are Kronecker deltas. It is the Kleisli category of D restricted to finite sets. **Conv** has as objects the simplices Δ_X and as morphisms the affine maps. Every kernel induces

$$T_K\colon \Delta_X \to \Delta_Y, \qquad T_K(p)(y) = \sum_x p(x)K(y \mid x). \tag{2}$$

Lemma 9.2 (Well-definedness and affineness)**.** *$T_K(p) \in \Delta_Y$, and $T_K(tp + (1-t)q) = tT_K(p) + (1-t)T_K(q)$.*

Proof. Every term is nonnegative, and $\sum_y T_K(p)(y) = \sum_x p(x) \sum_y K(y \mid x) = 1$. Affineness holds because (2) is linear in p. □

Lemma 9.3 (Identities and composition)**.** *$T_{I_X} = \mathrm{id}$ and $T_{L\circ K} = T_L \circ T_K$.*

Proof. $T_{I_X}(p)(x') = \sum_x p(x)\delta_{xx'} = p(x')$. Also $T_{L\circ K}(p)(z) = \sum_x p(x) \sum_y L(z \mid y)K(y \mid x) = \sum_y T_K(p)(y)L(z \mid y)$. □

Theorem 9.4 (Fully faithful functor)**.** *$T\colon$ **FinStoch** $\to$ **Conv** is a fully faithful functor. Each T_K is continuous and ℓ^1-non-expansive: $\|T_K(p) - T_K(q)\|_1 \leq \|p - q\|_1$.*

Proof. Functoriality is lemmas 9.2 and 9.3. An affine map of simplices is determined by the images of the vertices, which may be any points, so the affine maps $\Delta_X \to \Delta_Y$ are exactly the stochastic matrices. This is the comparison functor from the Kleisli to the Eilenberg–Moore category, restricted to free algebras. Non-expansiveness: $\sum_y |\sum_x (p-q)(x)K(y \mid x)| \leq \sum_x |(p-q)(x)|$ (checks E1, E2). □

Relation to the tower. T sends B to Δ_B and a kernel to an affine map, and a Beck diagram of kernels commutes in **FinStoch** iff it commutes in **Conv**, because T is faithful. Two cautions apply.

- The iterated levels $\mathsf{D}^2B, \mathsf{D}^3B$ are infinite-dimensional convex sets, not finite simplices.
- The shadows are not affine (proposition 5.1(c)), so they are not in the image of T, and their interchange is not a commutation of affine maps.

10 A Gaussian model: fusion and its type-correct domain

Let $B = \{A, B, C\}$ with uniform prior π. Let X_L, X_R be real channels, conditionally independent given the class $H = k$:

$$X_L \mid H = k \sim N(\mu_{L,k}, 1), \qquad X_R \mid H = k \sim N(\mu_{R,k}, 1),$$

with $\mu_{L,A} = -2$, $\mu_{L,B} = 0$, $\mu_{L,C} = 2$, $\mu_{R,A} = \mu_{R,C} = -1$ and $\mu_{R,B} = 1$. The left channel separates A from C; the right channel cannot. The level-one posteriors $Q_L(x)$, $Q_R(y)$ and $Q_{LR}(x,y)$ take values in Δ_B.

Proposition 10.1 (Fusion)**.** *$Q_{LR}(x,y) = \mathrm{Fuse}(Q_L(x), Q_R(y))$, where $\mathrm{Fuse}(q,r)_k = \dfrac{q_k r_k/\pi_k}{\sum_j q_j r_j/\pi_j}$.*

Proof. $Q_{LR} \propto \pi_k \ell_L(x \mid k)\ell_R(y \mid k)$, and $Q_L Q_R/\pi \propto \pi_k \ell_L(x \mid k)\ell_R(y \mid k)$. This holds exactly under conditional independence (check G1, discrete channels, exact). □

Domain. Fuse is defined on $U = \{(q, r) : \sum_j q_j r_j / \pi_j > 0\}$. Gaussian likelihoods are strictly positive, so $Q_L(x)$ and $Q_R(y)$ lie in the open simplex, and (Q_L, Q_R) maps all of $\mathbb{R}^2$ into U. The fusion square

$$\begin{array}{ccc} \mathbb{R}^2 & \xrightarrow{(Q_L,Q_R)} & U \\ & \searrow^{Q_{LR}} & \downarrow \mathrm{Fuse} \\ & & \Delta_B \end{array}$$

is therefore total and commutes everywhere. No restriction of the domain is needed. The pushforward $(Q_L, Q_R)_* P_{XY}$ has support touching the boundary of $\Delta_B \times \Delta_B$ only in closure, and the boundary has measure zero. The level-two laws are $S_L = (Q_L)_* P_X$, $S_R = (Q_R)_* P_Y$ and $S_{LR} = (Q_{LR})_* P_{XY}$, and the pushforward of the square along D commutes.

Shadows in the Gaussian model. By proposition 3.9, the shadows that commute are $E_{\sigma(X_L,H)}$ and $E_{\sigma(X_R,H)}$, with product $E_{\sigma(H)}$. The shadows $E_{\sigma(X_L)}$ and $E_{\sigma(X_R)}$ do not commute: their meet is trivial, because the joint density is positive, and X_L, X_R are dependent through H. A relabelling of the latent classes is a measure-preserving bijection that pulls back all three σ-algebras, so both commuting shadows are natural along it (proposition 4.6).

What the model shows. It gives a finite-dimensional, measurable pair of commuting shadows, a total fusion square, and a non-identity natural symmetry. It does not construct the Beck comparison maps. Nor does it show interchange for shadows that omit H; that interchange is false.

11 Positive tropicalisation and the Berkovich factorisation

11.1 Why a real non-Archimedean field

A non-Archimedean absolute value cannot restrict to the usual one on $\mathbb{R}$, so real probability vectors have no informative Berkovich points. For geometry to appear, probabilities must carry *scale*: families $p(t)$ whose coordinates behave like $c\,t^a$ as $t \to 0$.

Definition 11.1. $K = \mathbb{R}((t^{\mathbb{R}}))$ is the field of real Hahn series $\sum_{a \in S} c_a t^a$ with S well ordered. It is real closed and complete for $\mathrm{val}(f) = \min S$, with value group $\mathbb{R}$. Write $|f| = e^{-\mathrm{val}\, f}$. f is *positive* if its leading coefficient is positive. The real Puiseux series form a subfield, on which all exact checks are carried out.

The one fact behind everything that follows is this:

$$f, g > 0 \implies \mathrm{val}(f + g) = \min(\mathrm{val}\, f, \mathrm{val}\, g), \qquad \mathrm{val}(fg) = \mathrm{val}\, f + \mathrm{val}\, g. \tag{3}$$

So $\mathrm{val}\colon (K_{>0}, +, \times) \to (\mathbb{R}, \min, +)$ is a semiring homomorphism. This is Maslov dequantisation, the Laplace principle at finite level. A positive K-point is a family of distributions together with its large-deviation rates, and val keeps the rates.

11.2 The tropical tower

Definition 11.2 (Two monads)**.** $\mathsf{D}_K X$ is the set of finitely supported $p\colon X \to K_{\geq 0}$ with $\sum p = 1$. $\mathsf{D}_{\mathbb{T}} X$ is the set of finitely supported $v\colon X \to \mathbb{R} \cup \{\infty\}$ with $\min v = 0$, with unit $\eta_{\mathbb{T}}(x) = (0$ at x, ∞ elsewhere) and multiplication $\mu_{\mathbb{T}}(V)(x) = \min_u (V(u) + u(x))$.

Theorem 11.3 (Valuation is a monad morphism)**.** $\mathsf{D}_{\mathbb{T}}$ *is a monad, and* $\mathrm{val}\colon \mathsf{D}_K \Rightarrow \mathsf{D}_{\mathbb{T}}$ *is a morphism of monads:*

$$\mathrm{val} \circ \eta = \eta_{\mathbb{T}}, \qquad \mathrm{val}_X \circ \mu_X = \mu_{\mathbb{T},X} \circ \mathrm{val}_{\mathsf{D}_{\mathbb{T}}X} \circ \mathsf{D}_K(\mathrm{val}_X).$$

Proof. $\mathsf{D}_{\mathbb{T}}$ is the normalised distribution monad of the semiring $(\mathbb{R} \cup \infty, \min, +)$, and $\min_x \min_u(V(u) + u(x)) = \min V = 0$. The map val lands in $\mathsf{D}_{\mathbb{T}}$ because $\sum p = 1$ forces $\min \mathrm{val}\, p = 0$ by (3). The multiplication square is (3) applied to $\sum_\rho \Pi(\rho)\rho(x)$; atoms merged by $\mathsf{D}_K(\mathrm{val})$ have summed weights, whose valuation is the minimum (checks T1, T2). □

Corollary 11.4 (The tropical tower)**.** val *maps coherent* K*-contexts to coherent tropical contexts,* $e_\perp$ *and* $e_\top$ *to their tropical counterparts, and pooling on a cell to the tropical cell barycentre* $\min_j(\mathrm{val}\, p_j + \mathrm{val}\, Z_j) - \min_j \mathrm{val}\, p_j$.

11.3 Fibres

Definition 11.5. For positive $\rho \in \mathsf{D}_K X$, the *k-fibre* $F^{(k)}_\rho$ is the set of positive decompositions $\rho = \sum_{i \le k} w_i r_i$. For $v \in \mathsf{D}_{\mathbb{T}} X$, the *tropical k-fibre* $F^{(k)}_{\mathbb{T},v}$ is the set of $(w, u_1, \dots, u_k)$ with $w \in \mathsf{D}_{\mathbb{T}}[k]$ finite, $u_i \in \mathsf{D}_{\mathbb{T}} X$, and $\min_i(w_i + u_i(x)) = v(x)$ for all x.

Theorem 11.6 (Fibre lifting)**.** $\mathrm{val}(F^{(k)}_\rho) = F^{(k)}_{\mathbb{T},\mathrm{val}\,\rho}$.

Proof. $\subseteq$ is theorem 11.3. For $\supseteq$, let $\alpha(x) = \{i : w_i + u_i(x) = v(x)\}$ be the attainers at x. For $i \notin \alpha(x)$, choose positive $q_i(x)$ with $\mathrm{val}\, q_i(x) = w_i + u_i(x) > v(x)$. The remainder $m(x) = \rho(x) - \sum_{i \notin \alpha(x)} q_i(x)$ has the leading term of $\rho(x)$; split it equally among the attainers. Then $\sum_i q_i(x) = \rho(x)$ exactly. Put $w_i^K = \sum_x q_i(x)$ and $r_i = q_i / w_i^K$. Then $\mathrm{val}\, w_i^K = w_i + \min u_i = w_i$ and $\mathrm{val}\, r_i = u_i$. Check T3 runs this construction on 300 random points, with 83 tied coordinates. □

Proposition 11.7. $F^{(k)}_{\mathbb{T},v}$ *is a finite union of polyhedra, indexed by assignment types* $\alpha\colon X \to 2^{[k]} \setminus \{\emptyset\}$*. It contains the images of the two ends of the classical fibre:* $e_\perp \mapsto (0; v)$ *and* $e_\top \mapsto (w_x = v(x); u_x = \eta_{\mathbb{T}}(x))$.

11.4 Tropical secants

Definition 11.8. A model $V = \varphi(K^d_{>0}) \subseteq \mathsf{D}_K X$ is *positively parametrised* if the coordinates of φ are ratios of polynomials with nonnegative integer coefficients. Its positive tropicalisation is $\mathrm{trop}^+ V = \varphi_{\mathbb{T}}(\mathbb{R}^d)$. The k-mixture is $\mathrm{Mix}_k(V) = \{\sum_{i \le k} w_i r_i\}$, the positive part of the k-th secant variety, and its tropical analogue is $\mathrm{Mix}^{\mathbb{T}}_k(L) = \{\min_i(w_i + u_i)\}$.

Theorem 11.9 (Mixtures tropicalise to tropical secants)**.** $\mathrm{val}(V) = \mathrm{trop}^+ V$ *and* $\mathrm{val}(\mathrm{Mix}_k V) = \mathrm{Mix}^{\mathbb{T}}_k(\mathrm{trop}^+ V)$*, exactly. For a toric model* V_A*,* $\mathrm{trop}^+ V_A$ *is the row space of* A*, normalised, and* $\mathrm{Mix}^{\mathbb{T}}_k$ *is Develin's tropical secant of a linear space [23].*

Proof. $\mathrm{val}\, \varphi(\theta) = \varphi_{\mathbb{T}}(\mathrm{val}\, \theta)$ by (3), and val maps $K^d_{>0}$ onto $\mathbb{R}^d$. For mixtures, apply theorem 11.3 and choose lifts with prescribed valuations. □

Over an algebraically closed field, cancellation can make $\mathrm{trop}(\mathrm{Sec}_k)$ larger than the tropical secant [23, 24]. On the positive part there is no discrepancy. For matrices, the positive tropicalisation of the nonnegative-rank-$\le k$ locus is the Barvinok rank-$\le k$ locus. For real semialgebraic sets the image of val is Alessandrini's logarithmic limit set [11].

Example 11.10 (Dimensions)**.** Generic Jacobian ranks (floating point, tolerance 10^{-6}, 40 points; checks T6, T7):

model	k	$\dim \mathrm{Mix}_k^{\mathbb{T}}$	$\dim \mathrm{Mix}_k$ (affine cone)
3×3 independence	1, 2, 3	5, 8, 9	5, 8, 9
$2 \times 2 \times 2$ independence	1, 2	4, 8	4, 8

In each case the tropical dimension certifies the classical one, in the spirit of [25].

11.5 The Berkovich square

$(\mathbb{A}^m_K)^{\mathrm{an}}$ is the space of multiplicative seminorms on $K[z]$ extending $|\cdot|$ [22]. A K-point y gives the rigid point $|f|_y = |f(y)|$. On the torus there are two maps:

- the skeleton section $s(v)$, the Gauss point $|\sum_m a_m z^m|_{s(v)} = \max_m |a_m| e^{-\langle m,v\rangle}$;
- the retraction $\tau(\xi) = (-\log |z_j|_\xi)_j$.

They satisfy $\tau s = \mathrm{id}$, and $\tau(y) = \operatorname{val} y$ on rigid points. Payne's theorem [35] identifies the analytification with the inverse limit of tropicalisations, and Jell, Scheiderer and Yu prove the real analogue for semialgebraic sets [29].

Theorem 11.11 (Berkovich square)**.** *Let $\Phi\colon \mathbb{A}^m \to \mathbb{A}^n$ be polynomial with integer coefficients, for instance the k-mixture map, and assume the residue characteristic is 0. Then:*

(a) $\tau \circ \Phi^{\mathrm{an}} \circ s = \Phi_{\mathbb{T}}$ on the skeleton;

(b) $\tau \circ \Phi^{\mathrm{an}} = \Phi_{\mathbb{T}} \circ \tau$ on positive rigid points, if Φ has nonnegative coefficients;

(c) $\tau \circ \Phi^{\mathrm{an}} \geq \Phi_{\mathbb{T}} \circ \tau$ on all rigid points, strictly exactly where leading terms cancel.

$$\begin{array}{ccc} F_\rho^{(k)} \subseteq (\mathit{params})_K^{>0} & \xrightarrow{\ \Phi\ } & \mathrm{Mix}_k(V) \subseteq \mathsf{D}_K X \\ \Big\downarrow{\scriptstyle \mathit{rigid}} & & \Big\downarrow{\scriptstyle \mathit{rigid}} \\ (\mathbb{A}^m)^{\mathrm{an}} & \xrightarrow{\ \Phi^{\mathrm{an}}\ } & (\mathbb{A}^n)^{\mathrm{an}} \\ {\scriptstyle s}\big(\big\uparrow{\scriptstyle \tau}\big\downarrow & & \Big\downarrow{\scriptstyle \tau} \\ F_{\mathbb{T},v}^{(k)} \subseteq \mathbb{R}^m & \xrightarrow[\ \Phi_{\mathbb{T}}\ ]{} & \mathrm{Mix}_k^{\mathbb{T}} \subseteq \mathsf{D}_{\mathbb{T}} X \end{array}$$

Proof. (a) $-\log |\Phi_j|_{s(v)} = \min_m \langle m, v\rangle$, because nonzero integers have absolute value 1 (check T4). (b) is (3). (c) is the ultrametric inequality. □

11.6 What the analogies do not give

A Berkovich point is a multiplicative seminorm, not a probability measure. A rational domain is an algebraic localisation, not conditioning. The Kleisli category is not a site without specified covers. A distributive law is not a structure sheaf. Shadows are conditional expectations, not Grothendieck topologies. Two constructions that tensor $C(\Delta_X)$ with a non-Archimedean field fail for definite reasons:

- $\mathbb{R}$ has no isometric embedding into such a field;

- $C(\Delta_X)$ is not affinoid;
- over $\mathbb{R}$, Gel'fand duality returns Δ_X and the functor T of theorem 9.4, with nothing new.

What *does* hold is theorem 11.11. Positivity, through (3), is the hypothesis that turns the analogy into a theorem.

12 Shadows at zero temperature and secant geometry

12.1 The posterior tropicalises to the hard assignment

Proposition 12.1. *For positive $\rho = \sum_i w_i r_i$, the posterior $P(i \mid x) = w_i r_i(x)/\rho(x)$ has* $\operatorname{val} P(i \mid x) = w_i^{\mathbb{T}} + u_i(x) - v(x) \geq 0$, *with equality iff i attains the minimum. For convergent series, its $t \to 0$ limit is* $\operatorname{lc}(w_i r_i(x))/\operatorname{lc}(\rho(x))$ *on the attainers and* 0 *elsewhere, where* lc *is the leading coefficient (check T5).*

So conditioning on the observation tropicalises to argmin (MAP) assignment. Nearest-atom pooling is a *zero-temperature shadow*, and its Voronoi walls are tie loci. The same mechanism gives k-means as the small-variance limit of Gaussian mixtures [31].

Proposition 12.2 (Walls)**.** *(a) $\mu_{\mathbb{T}}$ is affine on each assignment cell. Crossing a tie hyperplane $w_i + u_i(x) = w_j + u_j(x)$ changes the assignment and the affine piece (check T8).*

(b) If $r_i = r_j$, the classical fibre contains the weight-split segment $\{w_i + w_j = c\}$. Its valuation image is the bent line $\{\min(w_i^{\mathbb{T}}, w_j^{\mathbb{T}}) = \operatorname{val} c\}$, made of two rays.

12.2 Statistical models as varieties

A parametric model is algebraic if $\theta \mapsto \mathbb{E}_\theta[s(X)]$ is polynomial or rational for a vector of sufficient statistics s. Its image is the *model variety*. For $N(\mu, \Sigma)$ on $\mathbb{R}^d$ with moments up to order r, the image is the Gaussian moment variety $G_{d,r}$, of dimension $d + d(d+1)/2$ when the parametrisation is generically injective. A mixture of k Gaussians has moment vector $\sum_i p_i m_r(\mu_i, \Sigma_i)$, so its image is the positive part of $\operatorname{Sec}_k(G_{d,r})$ [10]. Its parameter count is

$$\dim \Theta_{d,k} = k\big(d + \tfrac{d(d+1)}{2}\big) + k - 1,$$

and its expected dimension is $\min\{k \dim G_{d,r} + k - 1,\ \text{ambient dimension}\}$. The model is *generically identifiable* when the mixture map is generically finite-to-one, which requires $\dim \operatorname{Mix}_k = \dim \Theta_{d,k}$. It is *defective* when Sec_k has smaller than expected dimension. Defectivity is a global property of Sec_k, not a locus.

Proposition 12.3 (Moment order and the tower)**.** *(a) The order-r moment map of a two-component univariate Gaussian mixture has generic Jacobian rank* $\min(r, 5)$. *At order two the single-Gaussian image $\{(m, v + m^2)\}$ is already open, so second moments cannot see mixtures, and local identifiability needs $r \geq 5$ (check D1).*

(b) Affine statistics are constant on barycentre fibres. Regrouping a mixture of mixtures leaves the flattened law and all its moments unchanged (check D2). Hence the moment image of every tower level equals that of its flattening. If every node has m children, level n lies in Mix_{m^n}, *and the hierarchy itself is invisible to secant varieties. It lives in the convex-order fibre (proposition 5.2).*

Proof. (a) is direct computation at a rational point. (b) Moments are integrals, which are affine in the measure, and μ preserves integrals of affine functions. □

12.3 Walls of a component collision

For $\nu_\varepsilon = p\,N(\mu,\Sigma) + (1-p)\,N(\mu+\varepsilon v,\Sigma)$ with $0<p<1$, $\Sigma \succ 0$ and $v \neq 0$, the two-component fibre over ν_ε is finite for $\varepsilon \neq 0$, up to label swap. At $\varepsilon = 0$ it becomes the segment $\{(p_1,p_2;\mu,\Sigma) : p_1+p_2=1\}$. The collision locus is $\mathrm{Mix}_{k-1} \subset \mathrm{Mix}_k$, where the k-component parametrisation has positive-dimensional fibres. Tropically, by proposition 12.2(b), the collision is the locus $u_i = u_j$, where the tropical fibre gains a bent line.

Conjecture 12.4 (Wall-crossing, proposed)**.** *For a path ν_t crossing the collision locus transversally at ν_0, the change in the fibre is a finite sum $\sum_\sigma c_\sigma F_\sigma(\nu_0)$ over the collision strata σ through ν_0, with combinatorial coefficients c_σ determined by the local structure of the collision. In the single-collision example, $c_\sigma = 1$ and F_σ is the weight-split segment.*

The formal template is the wall-crossing of enumerative geometry, where the difference across a wall is a sum over strata. This is an analogy, not an established correspondence. The tropical picture of proposition 12.2 offers a concrete first test: the change in the tropical fibre across a tie or collision locus is a polyhedral computation.

Open problem 12.5 (Tropical identifiability)**.** For $\rho \in \mathrm{Mix}_k(V)$, is $\mathrm{val}(F_\rho^{(k)} \cap D_k(V)) = F^{(k)}_{\mathbb{T},\mathrm{val}\,\rho} \cap D_k^{\mathbb{T}}(\mathrm{trop}^+ V)$? For $V = \mathsf{D}_K X$ this is theorem 11.6. Lifting failures inside V would be tropical certificates of non-identifiability.

Open problem 12.6 (Order on tropical fibres)**.** Is the transitive closure of tropical pooling a partial order on $F_{\mathbb{T},v}$, with the two points of proposition 11.7 as bottom and top, and does it orient the tropical Beck defects of λ^σ as proposition 5.4 does classically?

Open problem 12.7 (Lifting obstructions as cohomology classes)**.** Sections 14 and 15 give two pieces of cohomology. One is the L^2 presheaf, whose Beck–Chevalley defect is measured by c. The other is the cohomology of tropical fibres, computed from assignment-cell nerves and determined by Farkas certificates. The Beck defects themselves are restrictions, not cocycles (proposition 14.7). The remaining question is whether lifting failures *inside a model* (open problem 12.5) define a coefficient system on the nerve whose H^1 is nonzero. That would make the tower's liftability certificates genuine cohomology classes.

13 The Hardy–Weinberg atlas

This section works one example through every layer of the paper. All the figures are produced by `atlas/atlas.py`, which computes each quantity from the formulas above; no data are sampled. The base is $X = \{0,1,2\}$, so a distribution is a point of a triangle. The model is the Hardy–Weinberg curve

$$r(\theta) = \big(\theta^2,\ 2\theta(1-\theta),\ (1-\theta)^2\big),$$

a positively parametrised toric curve, $r = (s^2, 2st, t^2)/(s+t)^2$. The tower uses three of its points, $\theta = 0.2, 0.5, 0.8$, with weights $\Pi = (0.35, 0.3, 0.35)$. An interactive version of the same figures accompanies the paper.

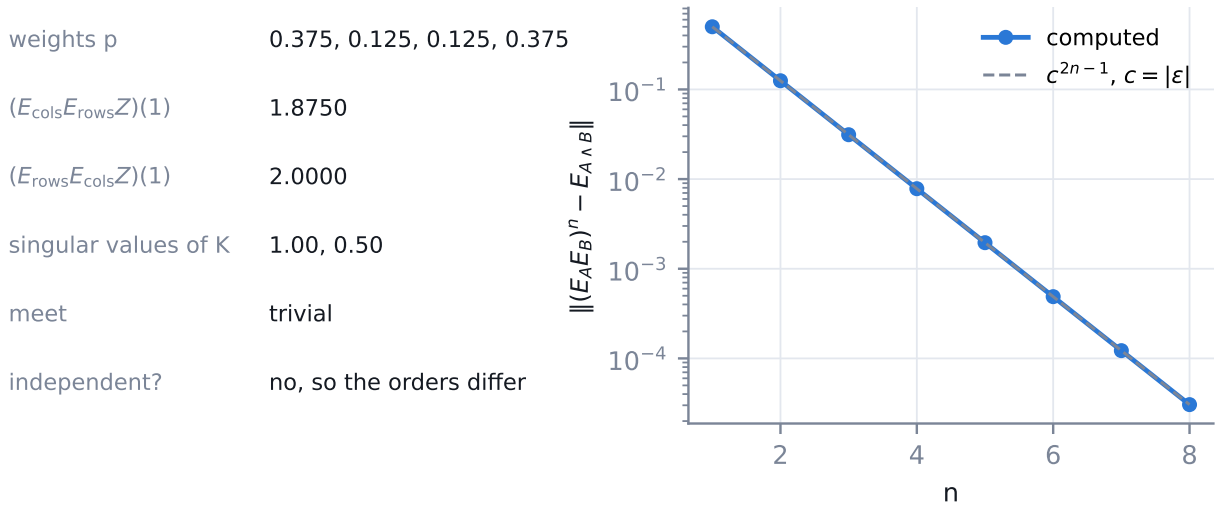


Figure 1: Algebra (theorems 3.3 and 3.6). Four points with weights $\frac{1\pm\varepsilon}{4}$, pooled by rows and by columns, at $\varepsilon = \frac{1}{2}$ (example 3.5). The two orders give 1.875 and 2.000 at point 1. The computed norms $\|(E_AE_B)^n - E_{A\wedge B}\|$ lie on c^{2n-1} with $c = \frac{1}{2}$, from 0.5 at $n = 1$ to 3.05×10^{-5} at $n = 8$.

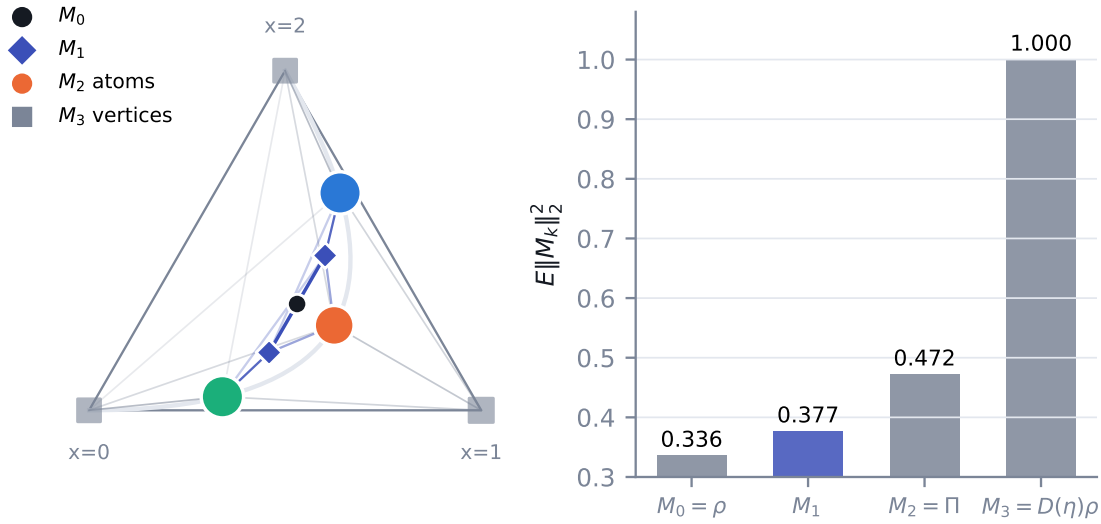


Figure 2: The tower as a martingale (proposition 5.2 and theorem 5.3). $M_0 = \rho$ (black), the level-two barycentres M_1 (diamonds), the atoms M_2 of Π (coloured, sized by mass) and the vertices M_3 (sized by ρ). Each position is a conditional expectation of the next. The energies $E\|M_k\|_2^2 = 0.336,\ 0.377,\ 0.472,\ 1.000$ increase along the convex-order chain. The outer pair, $\delta_\rho = e_\perp\Pi$ and $\mathsf{D}(\eta)\rho = e_\top\Pi$, are the two sides of the shift law's unit defect (proposition 5.4).

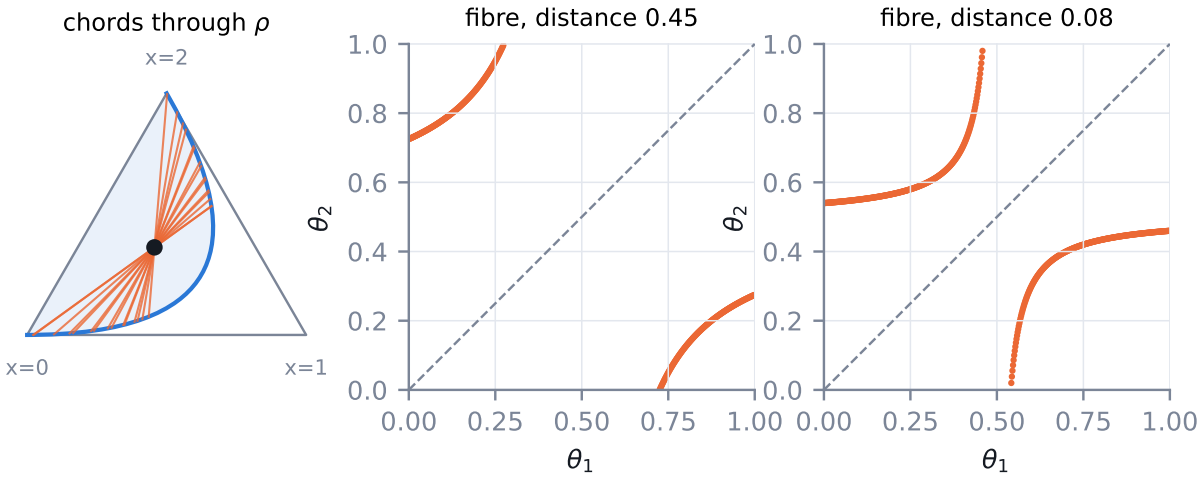


Figure 3: Secant geometry (section 12). Two-component mixtures of the curve fill its convex hull. The fibre over ρ is the family of chords through ρ, drawn as a curve in (θ_1, θ_2) coordinates. It has two branches, exchanged by relabelling the components. As ρ approaches the curve (distance $0.45 \to 0.08$), the branches close in on the diagonal $\theta_1 = \theta_2$, the collision locus where the fibre becomes the weight-split segment (proposition 12.2(b)).

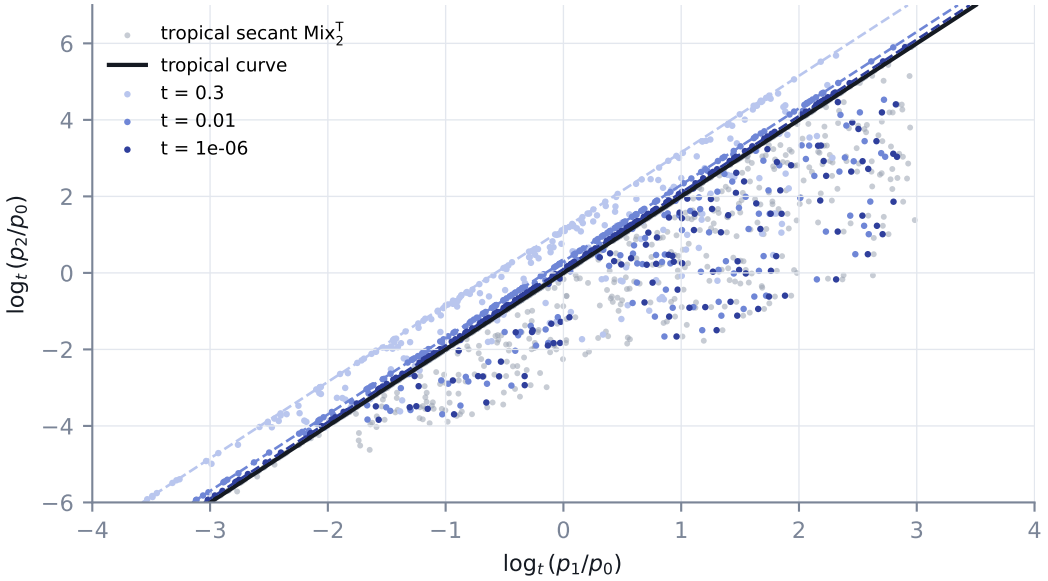


Figure 4: The Berkovich retraction in the chart $(\log_t(p_1/p_0), \log_t(p_2/p_0))$ (theorems 11.3, 11.9 and 11.11). Grey: the tropical secant $\min(w_1 + u_1, w_2 + u_2)$. Black: the tropical curve, which is the row space of the design. Coloured: the same 260 two-component mixtures at scales $t = 0.3$, 10^{-2}, 10^{-6}, and the image of the curve (dashed, shifted by $\log_t 2$). The largest coordinate deviation from the tropical limit is 1.60, 0.45 and 0.13. It decays like $1/\log(1/t)$, because only the leading-coefficient corrections $\log_t c$ remain.

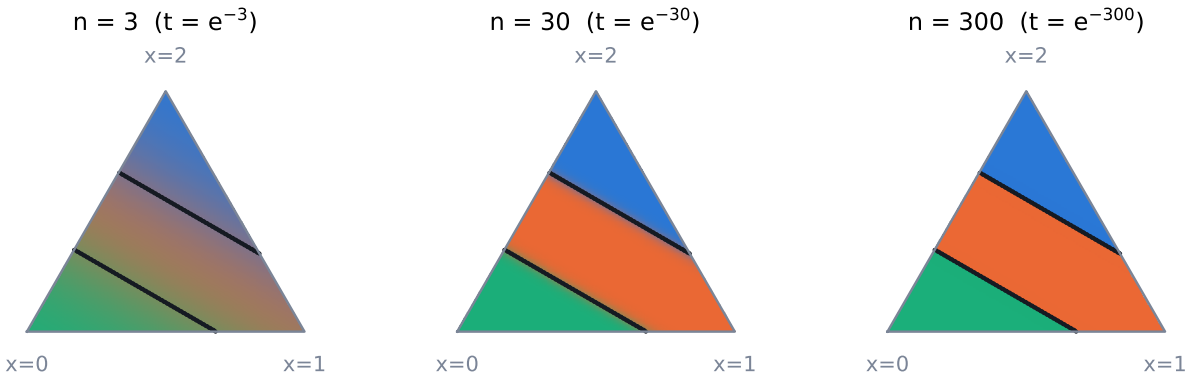


Figure 5: Zero-temperature shadows (propositions 12.1 and 12.2). Each empirical distribution q of n observations is coloured by the posterior over the three atoms, $\propto \Pi_i \exp(n \sum_x q(x) \log r_i(x))$. As n grows, which is $t = e^{-n} \to 0$, the colours harden into cells. The black walls are the $n \to \infty$ tie loci; they are straight because log-likelihoods are linear in q. The limiting map is nearest-atom pooling, discontinuous exactly on the walls (proposition 5.5).

14 Cohomology I: the L^2 presheaf and its Beck–Chevalley condition

Conditional expectations are the restriction maps of a presheaf, and the commutation theorem is its Beck–Chevalley condition. This section makes both statements precise and measures the failure.

Definition 14.1 (The L^2 presheaf)**.** Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, and let $\mathcal{S}$ be the poset of sub-σ-algebras of $\mathcal{F}$ (completed), ordered by inclusion. The presheaf $\mathcal{H}$ on $\mathcal{S}$ is defined by

$$\mathcal{H}(G) = L^2(G), \qquad \rho_{G'G} = E_G|_{L^2(G')} \colon \mathcal{H}(G') \to \mathcal{H}(G) \quad (G \subseteq G').$$

Each restriction has a section, the inclusion $\iota_{GG'} \colon L^2(G) \hookrightarrow L^2(G')$, with $\rho_{G'G}\iota_{GG'} = \mathrm{id}$.

Lemma 14.2. *$\mathcal{H}$ is a presheaf: $\rho_{GG} = \mathrm{id}$ and $\rho_{G'G}\rho_{G''G'} = \rho_{G''G}$ for $G \subseteq G' \subseteq G''$.*

Proof. This is the tower property $E_G E_{G'} = E_G$. □

Definition 14.3 (Beck–Chevalley)**.** A commuting square of inclusions $C \subseteq A$, $C \subseteq B$, $A \subseteq D$, $B \subseteq D$ satisfies the *Beck–Chevalley condition* for $\mathcal{H}$ if

$$\rho_{DA} \circ \iota_{BD} = \iota_{CA} \circ \rho_{BC} \colon L^2(B) \to L^2(A),$$

that is, $E_A f = E_C f$ for every $f \in L^2(B)$. Its *defect* is the operator $\beta_{A,B} = (\rho_{DA}\iota_{BD} - \iota_{CA}\rho_{BC})$ on $L^2(B)$.

Theorem 14.4 (Beck–Chevalley is conditional independence)**.** *Let A, B be sub-σ-algebras (completed), $C = A \wedge B = A \cap B$, and $D \supseteq A \vee B$. The following are equivalent:*

(i) the Beck–Chevalley condition holds for the square $C \subseteq A, B \subseteq D$;

(ii) $E_A E_B = E_{A \wedge B}$ (condition (ii) of theorem 3.3);

(iii) $E_A E_B = E_B E_A$;

(iv) A and B are conditionally independent given $A \wedge B$.

In general the defect satisfies

$$\|\beta_{A,B}\| = \|E_A E_B - E_{A\wedge B}\| = c,$$

where c is the cosine of the Friedrichs angle between $L^2(A)$ and $L^2(B)$. For finite partitions, c is the largest singular value below 1 of the normalised contingency matrix K. The condition is symmetric in A and B.

Proof. Write $M = L^2(A) \cap L^2(B)$.

Step 0: $M = L^2(C)$. The inclusion $\supseteq$ is clear. Conversely, let f have an A-measurable version f_A and a B-measurable version f_B. For rational q, the sets $\{f_A > q\} \in A$ and $\{f_B > q\} \in B$ differ by a null set. Both σ-algebras are complete, so each set lies in both, and f_A is $A \cap B$-measurable.

Step 1: (i)⇔(ii). By definition, (i) says $E_A g = E_C g$ for all $g \in L^2(B)$, that is, $E_A E_B = E_C E_B$. Since $C \subseteq B$, the tower property gives $E_C E_B = E_C$.

Step 2: (ii)⇒(iii). E_C is self-adjoint, so $E_B E_A = (E_A E_B)^* = E_C^* = E_C = E_A E_B$.

Step 3: (iii)⇒(ii). Put $P = E_A E_B$. Commutation gives $P^2 = E_A^2 E_B^2 = P$ and $P^* = E_B E_A = P$, so P is an orthogonal projection. Its range lies in $L^2(A)$, and since $P = E_B E_A$ it also lies in $L^2(B)$, hence in M. Every $f \in M$ has $Pf = f$. So P is the projection onto $M = L^2(C)$, which is E_C.

Step 4: (ii)⇒(iv). Let $U \in L^\infty(A)$, $V \in L^\infty(B)$ and $W \in L^\infty(C)$. Because $UW \in L^2(A)$ and $E_A V = E_A E_B V = E_C V$,

$$\mathbb{E}[UVW] = \langle UW, E_A V\rangle = \langle UW, E_C V\rangle = \mathbb{E}[U\,W\,E_C V] = \mathbb{E}[E_C U \cdot E_C V \cdot W],$$

where the last step conditions on C, using that $W\,E_C V$ is C-measurable. Hence $\mathbb{E}[UV \mid C] = E_C U \cdot E_C V$.

Step 5: (iv)⇒(ii). For bounded $U \in L^2(A)$ and $V \in L^2(B)$, $\langle U, E_A V\rangle = \mathbb{E}[UV] = \mathbb{E}[E_C U\,E_C V] = \langle U, E_C V\rangle$. Bounded functions are dense in $L^2(A)$, so $E_A V = E_C V$. Both sides are continuous in V and bounded V are dense in $L^2(B)$, so $E_A E_B = E_C E_B = E_C$.

Step 6: the norm. Split $g \in L^2(B)$ as $g = g_M + g_\perp$ with $g_M \in M$ and $g_\perp \in L^2(B) \ominus M$. Then $\beta g_M = g_M - g_M = 0$ and $E_C g_\perp = 0$, so $\beta g = E_A g_\perp$. Moreover $E_A g_\perp \perp M$, since $\langle E_A g_\perp, m\rangle = \langle g_\perp, m\rangle = 0$ for $m \in M$. Hence

$$\|\beta\| = \sup\big\{|\langle f, g\rangle| :\ f \in L^2(A) \ominus M,\ g \in L^2(B) \ominus M,\ \|f\| = \|g\| = 1\big\},$$

which is the definition of the Friedrichs cosine [33]. Also $E_A E_B - E_C = (E_A - E_C)E_B = \beta E_B$, and E_B maps the unit ball of $L^2(D)$ onto that of $L^2(B)$, so $\|E_A E_B - E_C\| = \|\beta\|$.

Step 7: the finite formula. In the bases $1_a/\sqrt{p(a)}$ and $1_b/\sqrt{p(b)}$, $E_A|_{L^2(B)}$ has matrix K. Its singular values are the cosines of the principal angles, and the value 1 belongs exactly to M (lemma 3.1). Removing M leaves the largest remaining singular value. Symmetry follows from (iv). □

Remark 14.5 (Reading the theorem)**.** The lattice of σ-algebras, or of partitions, which are assignments of outcomes to cells, carries the presheaf $\sigma \mapsto L^2(\sigma)$. Its restriction maps are conditional expectations. Two restrictions interchange exactly when the square through the meet is Beck–Chevalley, and the size of the failure is one number, the Friedrichs cosine. The same number is the exact rate of alternating projections (theorem 3.6), so it measures both how far the square is from commuting and how fast iterated restriction forgets the difference.

Check H1 verifies the equivalence on 300 random squares (199 conditionally independent, 101 not), and H2 verifies $\|\beta\| = c$.

14.1 Gluing and the degree-one obstruction

For a cover $\{A, B\}$ of D with overlap $C = A \wedge B$, form the Čech-type complex

$$0 \to L^2(D) \xrightarrow{\delta^0} L^2(A) \oplus L^2(B) \xrightarrow{\delta^1} L^2(C) \to 0,$$

with $\delta^0 f = (E_A f, E_B f)$ and $\delta^1(f_A, f_B) = E_C f_A - E_C f_B$. A pair in $\ker \delta^1$ is *compatible*: it agrees on the overlap. A *gluing* of a compatible pair is an f with $\delta^0 f = (f_A, f_B)$. The degree-one cohomology is $\check{H}^1 = \ker \delta^1 / \operatorname{im} \delta^0$.

Theorem 14.6 (Gluing)**.** *(a) $\delta^1 \delta^0 = 0$, and δ^1 is surjective.*

(b) The closure *of the image is the kernel: $\overline{\operatorname{im} \delta^0} = \ker \delta^1$, so the compatible pairs are the closure of the gluable pairs. The image itself equals $\ker \delta^1$ iff it is closed, iff $L^2(A) + L^2(B)$ is closed, iff $c < 1$; then $\check{H}^1 = 0$. This always holds in finite dimensions, so every compatible pair glues.*

(c) If $c = 1$, the image is dense in $\ker \delta^1$ but not closed. Then $\check{H}^1 = \ker \delta^1 / \operatorname{im} \delta^0$ is nonzero and non-Hausdorff, while the reduced cohomology $\ker \delta^1 / \overline{\operatorname{im} \delta^0}$ vanishes: some compatible pairs are limits of gluable pairs without being gluable.

(d) Under Beck–Chevalley the gluing is canonical: $f = f_A + f_B - E_C f_A$. This formula glues every compatible pair iff Beck–Chevalley holds.

(e) Gluings are not unique: $\ker \delta^0 = L^2(D) \ominus \big(L^2(A) + L^2(B)\big)$, which is nonzero unless $L^2(D) = \overline{L^2(A) + L^2(B)}$. So $\mathcal{H}$ satisfies the existence half of the sheaf condition (when $c < 1$), but not the uniqueness half.

Proof. (a) $E_C E_A = E_C = E_C E_B$, and $\delta^1(f, 0) = E_C f$ is onto $L^2(C)$. (b) The adjoint is $(\delta^0)^*(u, v) = u + v$, with kernel $\{(u, -u) : u \in L^2(A) \cap L^2(B) = L^2(C)\}$. So $\overline{\operatorname{im} \delta^0} = \{(f_A, f_B) : f_A - f_B \perp L^2(C)\} = \ker \delta^1$. By the closed range theorem, $\operatorname{im} \delta^0$ is closed iff $\operatorname{im}(\delta^0)^* = L^2(A) + L^2(B)$ is closed, which holds iff $c < 1$ [33]. (c) A dense, non-closed range is a proper subspace of its closure. (d) If Beck–Chevalley holds, $E_A f = f_A + E_C f_B - E_C f_A = f_A$, and symmetrically for B. Conversely, the pair $(f_A, E_C f_A)$ is compatible, and the formula returns $f = f_A$, which glues it iff $E_B f_A = E_C f_A$. Requiring this for all f_A is Beck–Chevalley. (e) $\ker \delta^0 = L^2(A)^\perp \cap L^2(B)^\perp$. □

Checks H3 and H4 verify (b) and (d) exactly on 300 squares.

Proposition 14.7 (Comparisons are restrictions, not gluing isomorphisms)**.** *Each Beck comparison computed in this paper (proposition 7.2) is a restriction $\rho_{G'G}$ with $G \subsetneq G'$. Such a map is surjective and not injective, and it strictly lowers the norm off $L^2(G)$. In particular no comparison map is an isomorphism, so the comparisons cannot be the gluing isomorphisms of a structure sheaf. What they are is the structure maps of $\mathcal{H}$. The interchange of two of them is the Beck–Chevalley condition (theorem 14.4), and its failure is measured by c.*

Proof. $\rho_{G'G} \iota_{GG'} = \mathrm{id}$ gives surjectivity, and $L^2(G') \ominus L^2(G) \neq 0$ is its kernel. Pythagoras gives the strict decrease (check H5). □

14.2 A unified coefficient system

The objects of this section and of section 16 fit into one coefficient system with two layers: a nonlinear layer of laws, and its linearisation, which is the L^2 presheaf. The layers capture the local

sections, their compatibility, the quantitative defect and the information content together; only positivity is left to convex duality.

Let $\mathcal{P}(\sigma)$ be the simplex of laws on the cells of σ, with marginalisation as restriction and KL divergence as its canonical divergence; this is the dually flat structure of information geometry [19]. Fix a full-support law P.

Proposition 14.8 (The L^2 presheaf is the tangent presheaf)**.** *For $v \in L^2_0(P)$ and $P_t = P(1+tv)$, the derivative at $t=0$ of $\log P_t(\text{cell of } \sigma)$ is $E_\sigma v$. So marginalisation differentiates to conditional expectation, and the tangent presheaf of $\mathcal{P}$ at P, with the Fisher metric, is $\sigma \mapsto L^2_0(\sigma)$ with restrictions E_σ.*

Proof. $\frac{d}{dt}\log\sum_{\omega\in K} P(\omega)(1+tv(\omega))\big|_{t=0} = \sum_{\omega\in K} P(\omega)v(\omega)/P(K) = E_\sigma v$ on the cell K. The Fisher metric at P is $\langle u, v\rangle = \mathbb{E}_P[uv]$ in these coordinates. Check I1. □

Theorem 14.9 (The Laplacian of the gluing complex)**.** *For two windows A, B with meet C, the degree-zero Laplacian of the gluing complex of theorem 14.6 is $(\delta^0)^*\delta^0 = E_A + E_B$. Its spectrum consists of 0 on $\ker\delta^0$, 2 on $L^2(C)$, 1 on the parts of $L^2(A)$ and $L^2(B)$ orthogonal to the other window, and the pairs $1 \pm \cos\theta_k$ for the principal angles θ_k between $L^2(A) \ominus L^2(C)$ and $L^2(B) \ominus L^2(C)$. Hence its smallest non-zero eigenvalue is $1-c$, and the non-trivial eigenvalues of $E_A E_B E_A$ are $\cos^2\theta_k$, which give the alternating-projection rate c^{2n-1} of theorem 3.6.*

Proof. $(\delta^0)^*(u,v) = u + v$, so $(\delta^0)^*\delta^0 f = E_A f + E_B f$. Decompose $L^2(D)$ by the canonical decomposition of two projections: on each principal plane $E_A + E_B$ acts as the sum of two rank-one projections at angle θ_k, with eigenvalues $1 \pm \cos\theta_k$; on $L^2(C)$ both projections are the identity; on the remaining parts of $L^2(A)$ and $L^2(B)$ one of them is; and on $L^2(A)^\perp \cap L^2(B)^\perp = \ker\delta^0$ both vanish. Since $1 - \cos\theta \le 1$, the smallest non-zero eigenvalue is $1 - \max_k \cos\theta_k = 1 - c$. On a principal plane $E_A E_B E_A$ is $\cos^2\theta_k$ times a rank-one projection. Check I2 on 300 random pairs of partitions. □

Remark 14.10 (One metric defect, three consequences)**.** For two contexts A, B with $C = A \wedge B$, the Friedrichs cosine c of $L^2(A) \ominus L^2(C)$ and $L^2(B) \ominus L^2(C)$ controls three things at once: the Beck–Chevalley defect, $\|\beta_{A,B}\| = c$ (theorem 14.4); the degree-zero spectral gap $1-c$ of the gluing complex (theorem 14.9); and the rate of alternating projections (theorem 3.6). A positive gap is equivalent to closed range of δ^0, which is equivalent to every compatible pair gluing (theorem 14.6(b)). The three regimes are:

- $c = 0$: $E_A E_B = E_{A\wedge B}$ in one step; this is conditional independence given the meet, and interchange is strict.
- $0 < c < 1$: the gap $1-c$ is positive, every compatible L^2 pair glues, and $\|(E_A E_B)^n - E_{A\wedge B}\| = c^{2n-1}$, so iterative reconciliation converges geometrically, with conditioning controlled by $1-c$.
- $c = 1$, which occurs only in infinite dimensions: the gap closes, the range of δ^0 is not closed, and some compatible pairs are limits of gluable pairs without being gluable themselves (theorem 14.6(c)). Alternating projections still converge for each vector, but with no uniform rate, and arbitrarily slowly [12].

In finite dimensions, which covers every example in this paper, only the first two regimes occur, and the gap measures how well conditioned the gluing is, not whether it happens. This is the sheaf-Laplacian picture of [21] for the presheaf $\mathcal{H}$. None of these linear statements decides whether compatible probability marginals admit a nonnegative global law; indeed the linear layer never

obstructs, since every compatible family has a signed global section (see the end of this subsection). Existence is the separate convex-feasibility problem of section 16, where failure needs a cycle in the cover (proposition 16.3).

Proposition 14.11 (KL is the global form of the defect)**.** *Let $P^* = P_{AC}P_{CB}/P_C$ be the Markov glue of the windows of P, and $\mathcal{M}$ the set of full-support laws with $A \perp B \mid C$.*

(a) For every $Q \in \mathcal{M}$, $\mathrm{KL}(P\|Q) = \mathrm{KL}(P\|P^*) + \mathrm{KL}(P^*\|Q)$. So P^* is the KL projection of P onto $\mathcal{M}$, and $\mathrm{KL}(P\|P^*) = I(A;B \mid C)$.

(b) $P \in \mathcal{M}$ iff $I(A;B \mid C) = 0$ iff $c = 0$ iff Beck–Chevalley holds (theorem 14.4).

(c) Near $\mathcal{M}$, $I(A;B \mid C) = \frac{1}{2}\sum_c P(c)\sum_k \sigma_k(c)^2\,(1 + o(1))$, where $\sigma_k(c)$ are the non-trivial singular values of the normalised table of (A,B) given $C = c$, that is, the principal cosines of theorem 14.9 within each block. Far from $\mathcal{M}$ this is only an approximation.

Proof. (a) $\log Q = \log Q_C + \log Q_{A|C} + \log Q_{B|C}$ for $Q \in \mathcal{M}$, and P and P^* have the same (A,C) and (C,B) margins, so $\mathbb{E}_P \log Q = \mathbb{E}_{P^*} \log Q$; the identity follows by adding and subtracting $\mathbb{E}_P \log P^*$, and $P^* \in \mathcal{M}$ (proposition 16.3(a)). This is the Pythagorean theorem for a KL projection onto a log-linear family [20]. (b) $\mathrm{KL}(P\|P^*) = 0$ iff $P = P^*$; the rest is theorem 14.4. (c) Within a block, mutual information is half the χ^2-information to second order, and the χ^2-information of a table is the sum of the squared non-trivial singular values of its normalised form. Checks I3 (500 laws) and I4: the ratio tends to 1 (maximal deviation 0.074, 0.020, 0.007, 0.001 at distance 0.3, 0.1, 0.03, 0.01) and ranges over $[0.75, 1.27]$ for random laws. □

What the system does not decide. Neither layer decides whether a positive global law exists. The tangent layer never obstructs: every compatible family of window tables has a signed global section [4], as proposition 17.1 shows in the three-token case. Existence is decided by linear programming, with certificates such as $\Phi \le 2$ and the cycle inequalities. Check I5 finds a signed section for all of 400 random compatible families on the triangle of binary variables and a positive one for only 287 of them.

captures	law layer $\mathcal{P}$	tangent layer L_0^2
local sections	laws on each context, marginalisation	$L_0^2(\sigma)$, conditional expectation (proposition 14.8)
compatibility	the Markov glue; the model $\mathcal{M}$	$E_A E_B = E_{A\wedge B}$ (theorem 14.4)
quantitative defect	$\mathrm{KL}(P\|P^*) = I(A;B \mid C)$	c; Laplacian gap $1 - c$ (theorem 14.9)
information	KL Pythagoras (proposition 14.11)	$\frac{1}{2}\sum \sigma_k^2$, to leading order
existence	LP duality and its certificates	never obstructed

15 Cohomology II: fibres as topology

Classical barycentre fibres are convex (proposition 5.2), so their topology is trivial. So are the tower's level-two lifting fibres, which are polytopes [1, Def. 10.14]. Nontrivial topology appears in two places: when components are constrained to a curved model, and in tropical fibres, which are unions of polyhedra. This section computes the second kind exactly.

15.1 The cover by assignment cells

For the Hardy–Weinberg model, a tropical two-component decomposition has parameters $\vartheta = (\beta, d_1, d_2) \in \mathbb{R}^3$. Here β is the weight difference, $w_1 = \max(0, -\beta)$, $w_2 = \max(0, \beta)$, and d_i is the rate of component i, so that by theorem 11.9 $u(d) = (0, d, 2d) - \min(0, 2d)$. The *tropical fibre* over a target $v \in \mathsf{D}_{\mathbb{T}}X$ is

$$F_v = \big\{\vartheta : \min_i \big(w_i + u(d_i)(x)\big) = v(x) \text{ for all } x\big\}.$$

Definition 15.1 (Assignment cells)**.** For $\alpha\colon X \to \{1, 2\}$ (a designated attainer for each outcome) and a sign pattern $\sigma \in \{\pm\}^3$ of (β, d_1, d_2), let

$$P_{\alpha,\sigma} = \big\{\vartheta \in \text{orthant } \sigma : \; w_{\alpha(x)} + u(d_{\alpha(x)})(x) = v(x), \quad w_j + u(d_j)(x) \geq v(x) \; \forall j, x\big\}.$$

Lemma 15.2. *Each $P_{\alpha,\sigma}$ is a closed convex polyhedron. The cells cover F_v, and every intersection of cells is a polyhedron.*

Proof. Within an orthant every term is affine in ϑ, so each cell is defined by finitely many affine equalities and inequalities. A point of F_v lies in the cell given by its orthant and by any choice of attainer at each x. □

Theorem 15.3 (Fibre cohomology from the nerve)**.** *Let N_v be the nerve of the cover $\{P_{\alpha,\sigma}\}$. Then $H^*(F_v; \mathbb{Q}) \cong \check{H}^*(\{P_{\alpha,\sigma}\}; \mathbb{Q}) \cong H^*(N_v; \mathbb{Q})$. A set of cells spans a face of N_v iff its joint system of affine constraints is feasible. By Farkas' lemma, it is* not *a face iff there is a certificate $(y_{\mathrm{eq}}, y_{\mathrm{ub}} \geq 0)$ with $A_{\mathrm{eq}}^{\top} y_{\mathrm{eq}} + A_{\mathrm{ub}}^{\top} y_{\mathrm{ub}} = 0$ and $b_{\mathrm{eq}} \cdot y_{\mathrm{eq}} + b_{\mathrm{ub}} \cdot y_{\mathrm{ub}} < 0$. So the cohomology of the fibre is determined by which assignment patterns are certified impossible.*

Proof. The nerve theorem holds for finite covers by closed convex sets in $\mathbb{R}^d$: every nonempty intersection is convex, hence contractible [32]. The face criterion is Farkas' lemma [34]. □

These certificates have the same form as the tower's lifting certificates [1, Thm. 10.8, Thm. 10.13(iii)]. There, a separating functional certifies that a claim has no admissible lift. Here, a Farkas vector certifies that a pattern of attainers has no tropical decomposition. Theorem 15.3 upgrades such certificates from yes/no answers to cohomological data: the complex of patterns that are *not* certified impossible computes the homology of the fibre.

15.2 Local lifts as sections

Proposition 15.4 (Local lifts)**.** *Work in the full simplex model $\mathsf{D}_K X$, and fix a positive ρ with* $\mathrm{val}\,\rho = v$.

(a) For each cell, the construction in the proof of theorem 11.6, with the remainder at x given to $\alpha(x)$, is a section of val *over the rational points of the cell.*

(b) Over a fixed tropical point, the set of lifts is convex. A convex combination of two exact decompositions of ρ with the same valuation pattern is again one, because leading coefficients of positive terms add without cancelling.

(c) Consequently, on every nonempty face of the nerve, local lifts from different cells are joined by a segment of lifts. The lifting data add no obstruction beyond the emptiness certificates of theorem 15.3.

Proof. (a) is theorem 11.6 with designated attainers. (b) holds because the decomposition equations are linear in $(q_i(x))$, and the valuation pattern is fixed by positivity (check K4). (c) follows from (b) on each common point. □

For a model V, lifting into V is open problem 12.5. A lifting failure inside V would be a new kind of certificate. Its interaction with the nerve is the natural next computation.

15.3 The computation for Hardy–Weinberg

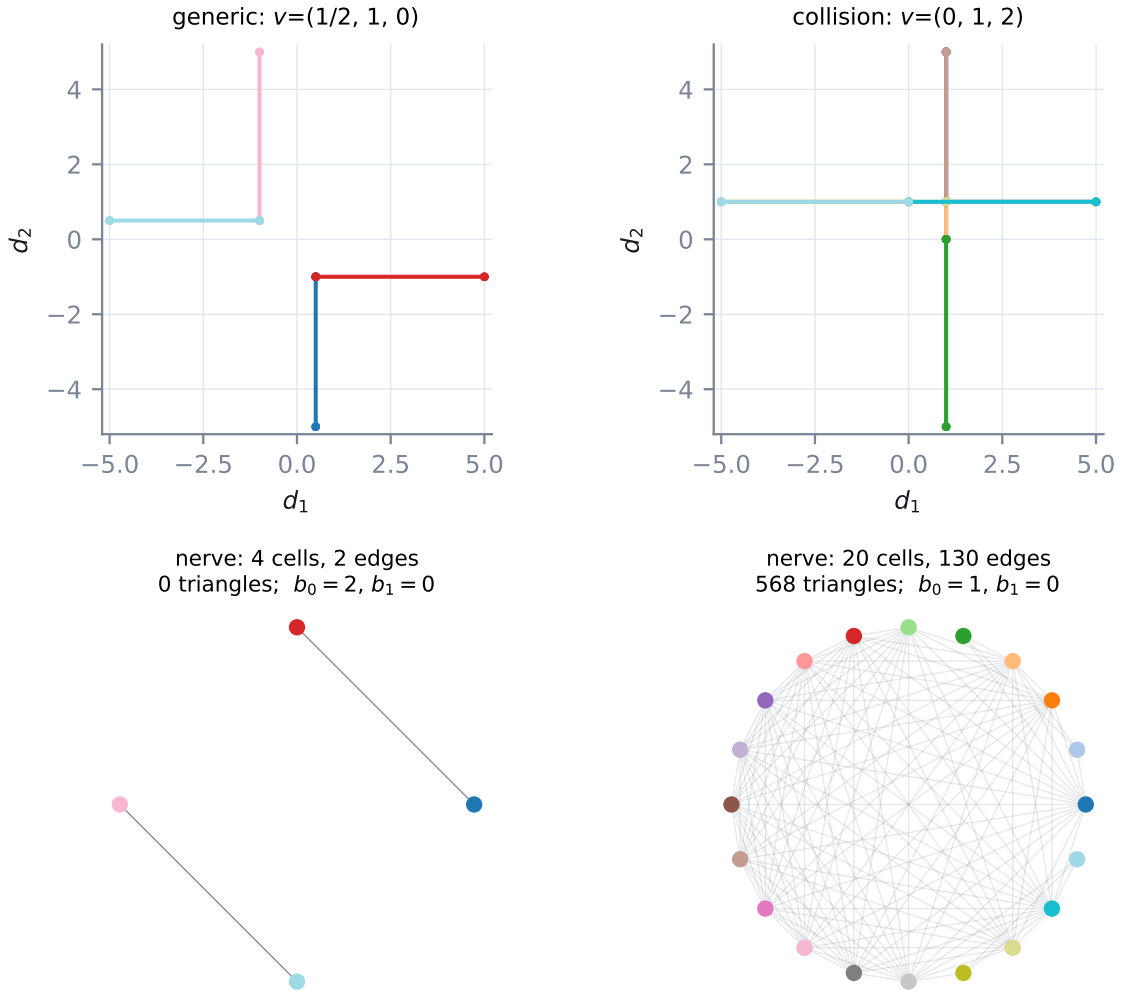


Figure 6: Tropical fibres F_v projected to (d_1, d_2), coloured by assignment cell, and their nerves. *Generic* $v = (\frac{1}{2}, 1, 0)$: four cells and two edges. There are two components, each a bent line, exchanged by swapping the components, and $b_0 = 2$, $b_1 = 0$. *Collision* $v = (0, 1, 2)$, on the tropical curve: twenty cells, 130 edges and 568 triangles. The branches are joined, and $b_0 = 1$, $b_1 = 0$.

Theorem 15.5 (Topology of the Hardy–Weinberg fibres)**.** *For the generic target* $v = (\frac{1}{2}, 1, 0)$*,* F_v *has Betti numbers* $(b_0, b_1) = (2, 0)$*. Its two components are exchanged by the label swap* $(\beta, d_1, d_2) \mapsto (-\beta, d_2, d_1)$*. For the collision target* $v = (0, 1, 2)$*, which lies on the tropical curve,* $(b_0, b_1) = (1, 0)$*.*

Proof. Exact computation (checks K1–K3). Every nonempty cell and face has an exact rational witness: 6 for the generic target and 718 for the collision target. Every empty cell and face among the candidates has an exact Farkas certificate: 64 and 104 respectively. The Betti numbers are ranks over $\mathbb{Q}$ of the coboundary matrices of the nerve. □

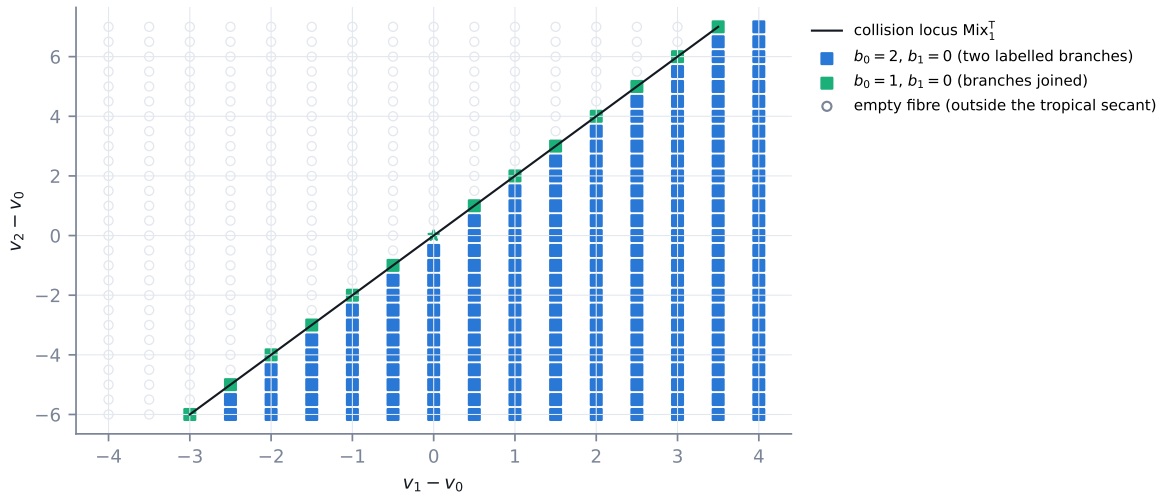


Figure 7: The homology of the fibre as a function of the target, over a grid of 17×27 targets in the chart $(v_1 - v_0, v_2 - v_0)$. Of these, 236 lie outside the tropical secant (empty fibre). On the 209 targets inside the secant and off the collision locus, $b_0 = 2$; on the 14 targets on the locus, $b_0 = 1$. $b_1 = 0$ wherever it was computed, which is every target except $v = (0, 0, 0)$, where all 64 cells are nonempty and only b_0 was computed. This is the homological form of the collision wall of section 12.

Reading. Over a generic target there are two continuous families of decompositions, one for each labelling. On the collision locus a component may coincide with the other, and the families merge. The same jump appears classically in fig. 3, where the two branches of the classical fibre meet the diagonal. Homology turns the proposed wall-crossing of conjecture 12.4 into a computable invariant. In this example the jump is $\Delta b_0 = -1$ across the collision locus, with no change in b_1. On the scanned grid, tie walls in the target (proposition 12.2(a)) change the cell structure of the fibre but not its homology.

16 Example: hallucination in a three-token model

This section connects the L^2 presheaf to the tower's notion of hallucination [1, §10.5]. A hallucination there is an *obstructed claim*: a statement that is admissible locally but has no coherent completion. We use the smallest model in which this happens, three token positions observed pairwise. Every number below is checked exactly by `check_hallucination.py`.

16.1 The model

Three binary token positions X_0, X_1, X_2 are seen through the contexts (windows) $\{0,1\}$, $\{1,2\}$ and $\{0,2\}$. In each context the two tokens agree with probability ε:

$$P_{X_iX_j} = \begin{pmatrix} \varepsilon/2 & (1-\varepsilon)/2 \\ (1-\varepsilon)/2 & \varepsilon/2 \end{pmatrix}.$$

The tables come from a mixture of sources $s \in \{A, B, U\}$, each with its own agreement ε_s. With weights w in the simplex, the pairwise tables mix linearly:

$$\varepsilon(w) = w_A\varepsilon_A + w_B\varepsilon_B + w_U\varepsilon_U.$$

We take $\varepsilon_A = 0.20$, $\varepsilon_B = 0.30$ and $\varepsilon_U = 0.40$.

16.2 Hallucination is a failure of gluing

Proposition 16.1 (Global sections and their certificate)**.** *(a) A joint law of (X_0, X_1, X_2) with these pairwise tables exists iff $\varepsilon \geq \frac{1}{3}$.*

(b) For $\varepsilon < \frac{1}{3}$, the functional $\Phi(P) = \sum_{i<j} P(X_i \neq X_j)$ certifies the obstruction. Every joint has $\Phi \leq 2$, because each of the eight joint outcomes has at most two disagreeing pairs. The pairwise tables give $\Phi = 3(1-\varepsilon)$, which exceeds 2 by exactly $1 - 3\varepsilon$.

(c) For $\varepsilon \geq \frac{1}{3}$, the global sections form the segment $J + s\chi$, $|s| \leq \min(a, b)$. Here $\chi(x) = (-1)^{x_0+x_1+x_2}$ spans the one-dimensional kernel of the pairwise-margin map, and J is the symmetric joint with $a = \frac{3\varepsilon-1}{4}$ on 000 and 111 and $b = \frac{1-\varepsilon}{4}$ on the other six outcomes.

Proof. (a) and (b): this is [2, Thm. 4.8] at $\tau = 0$. All three correlations are $E_{ij} = 2\varepsilon - 1$, and the 3-cycle inequality with all signs negative reads $-3(2\varepsilon - 1) \leq 1$. The bound $\Phi \leq 2$ is checked atom by atom. (c) The margin map has rank 7 on $\mathbb{R}^8$, and χ lies in its kernel. J reproduces the tables. Nonnegativity of $J + s\chi$ gives the range of s (checks L1, L2). □

In the language of [1, §10], each pairwise table is an admissible local claim, and for $\varepsilon < \frac{1}{3}$ the family is an obstructed claim. Its certificate is the test function Φ, a separating functional of the kind in [1, Thm. 10.8]. The parity χ is not a certificate. It indexes the non-uniqueness of the joint when one exists.

16.3 The wall in weight space, and a token stream

By proposition 16.1, the mixture is liftable iff $\varepsilon(w) \geq \frac{1}{3}$. This is a half-space in the weight simplex: the wall $\varepsilon(w) = \frac{1}{3}$ is a straight line, and the liftable side is the side of the coherent source U (the only source with $\varepsilon_s \geq \frac{1}{3}$). In the coordinates $\theta = w_A/(w_A + w_B)$ and $\mu = w_U$, the wall becomes the curve $\mu(\theta) = (\frac{1}{3} + \theta)/(1 + \theta)$. The liftable side is $\mu \geq \mu(\theta)$: more weight on the coherent source restores a joint.

For the 12-step stream $(\theta_t, \mu_t) = (0.10, 0.10), (0.20, 0.15), \ldots, (0.30, 0.90)$ of fig. 8, the agreement ε_t rises from 0.301 to 0.387. The stream is obstructed (hallucinating) for $t = 1, \ldots, 7$, with $\varepsilon_7 = 0.332$, and liftable from $t = 8$, with $\varepsilon_8 = 0.337$. The transition is the crossing of the wall between steps 7 and 8 (check L6). This wall is classical: it is linear in the weights, and it is not a tropical wall (section 16.6).

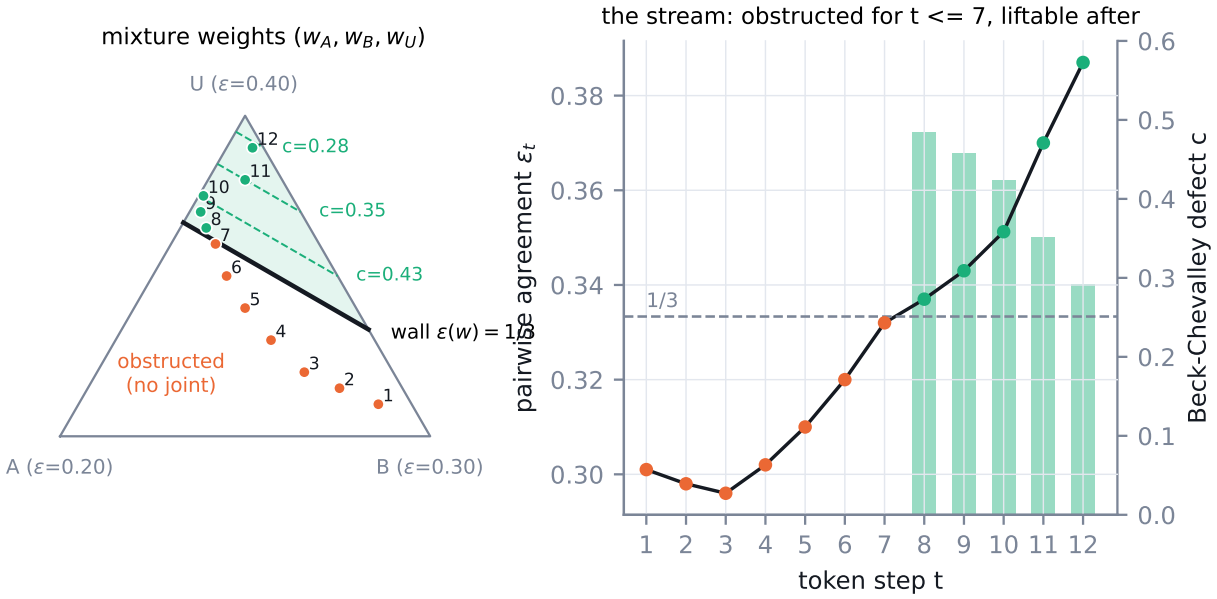


Figure 8: The three-token hallucination example. Left: the weight simplex. The straight wall $\varepsilon(w) = \frac{1}{3}$ separates the obstructed region (no joint law) from the liftable region (shaded). Dashed lines are level sets of the Beck–Chevalley defect c of the symmetric joint, and points are the 12 stream steps (orange obstructed, green liftable). Right: ε_t along the stream, which crosses $\frac{1}{3}$ between $t = 7$ and $t = 8$, and the defect c_t (bars) once a joint exists.

16.4 The L^2 presheaf on context windows

Once the claim is liftable, choose a full-support global section P (for instance J, for $\frac{1}{3} < \varepsilon < 1$). This gives a probability space, and the windows give two σ-algebras on it:

$$A = \sigma(X_0, X_1), \qquad B = \sigma(X_1, X_2).$$

Their shadows E_A and E_B are what a reader confined to one window can compute.

Proposition 16.2 (Window shadows)**.** *For a full-support section P:*

(a) the meet is $A \wedge B = \sigma(X_1)$, the shared token;

(b) the window square satisfies Beck–Chevalley iff $X_0 \perp X_2 \mid X_1$ under P, that is, iff the sequence is Markov through the shared token;

(c) the defect is $c = \max_{x_1} \left|\operatorname{corr}_P(X_0, X_2 \mid X_1 = x_1)\right|$, and alternating the two window shadows converges to $E[\cdot \mid X_1]$ at rate c^{2n-1};

(d) at the symmetric section J, $c = \dfrac{|2\varepsilon - 1|}{2\varepsilon}$;

(e) a Markov section exists in the segment $J + s\chi$ iff $\varepsilon = \frac{1}{2}$ (for $\frac{1}{3} < \varepsilon < 1$). On 99 sections across the segment, the defect is smallest at the symmetric section.

Proof. (a) The cells of A are values of (x_0, x_1) and those of B values of (x_1, x_2). Two cells meet iff they share x_1, so the components of $G(A, B)$ are the two values of x_1 (lemma 3.1). (b) This

is theorem 14.4(iv) with meet $\sigma(X_1)$. (c) K is block diagonal over the two components. Each block is the normalised 2×2 table of (X_0, X_2) given $X_1 = x_1$, whose singular values are 1 and the absolute correlation. Now apply theorems 3.6 and 14.4. (d) Given $X_1 = 0$, the table of (X_0, X_2) is proportional to $\left(\begin{smallmatrix} a & b \\ b & b \end{smallmatrix}\right)$. Its correlation is $(a-b)/(2(a+b)) = (2\varepsilon - 1)/(2\varepsilon)$, and the case $X_1 = 1$ is symmetric. (e) Under the Markov property, $P(X_0 = X_2 \mid X_1) = \varepsilon^2 + (1-\varepsilon)^2$. This must equal the table value ε, so $\varepsilon \in \{\frac{1}{2}, 1\}$. The minimisation is computed (checks L3–L5, L7). □

Reading. Two context windows that share a token commute as conditional expectations exactly when everything the first token says about the third passes through the shared token. The defect c is the long-range coupling that no single window represents, and c^{2n-1} is how fast repeated window-by-window conditioning forgets it. Along the stream, once a joint exists the defect falls from $c_8 = 0.484$ to $c_{12} = 0.292$ as weight moves to the coherent source. It never reaches 0, because $\varepsilon = \frac{1}{2}$ is not attainable with these sources.

16.5 Three layers of obstruction

regime	structure	certificate or measure
$\varepsilon < \frac{1}{3}$	no global section (hallucination)	$\Phi > 2$, by $1 - 3\varepsilon$ (proposition 16.1)
$\frac{1}{3} \le \varepsilon < 1$, $\varepsilon \neq \frac{1}{2}$	sections form a segment; windows do not commute	Beck–Chevalley defect $c > 0$ (proposition 16.2)
$\varepsilon = \frac{1}{2}$	a Markov section exists	$c = 0$ at that section

The first layer is a failure of gluing for the presheaf of distributions on the context cover. The second is the Beck–Chevalley defect of the L^2 presheaf on the glued space. The tower's obstructed claims live in the first layer. The L^2 presheaf measures what remains once the claim is coherent.

The two layers are logically independent: a gluing failure does not imply a Beck–Chevalley defect, nor the reverse. The next proposition makes this exact. Gluing failure comes from cycles in the cover; the defect comes from coupling inside the glued space.

Proposition 16.3 (Two windows always glue; failure needs a cycle)**.** *(a) Let U, V, W be finite random variables, and let P_{UV} and P_{VW} be full-support laws with the same V-marginal P_V. Then*

$$P(u, v, w) = \frac{P_{UV}(u, v)\, P_{VW}(v, w)}{P_V(v)}$$

is a joint law with these two windows. It is the only one in which $U \perp W \mid V$, that is, the only one for which the windows $\sigma(U, V)$ and $\sigma(V, W)$ satisfy Beck–Chevalley. Every other gluing violates $U \perp W \mid V$. So a cover by two windows never fails to glue.

(b) More generally, every compatible family of marginals on a cover glues whenever the cover is acyclic, that is, its windows can be removed one at a time so that each removed window meets the rest inside a single remaining window [15]. The three pair windows of proposition 16.1 are not acyclic: their nerve is a hollow triangle ($b_1 = 1$; check M8), and proposition 16.1 exhibits the failure for $\varepsilon < \frac{1}{3}$.

(c) The defect does not detect the failure. As $\varepsilon \downarrow \frac{1}{3}$ the fibre shrinks to the single section J, where $c = |2\varepsilon - 1|/(2\varepsilon) = \frac{1}{2}$. So at the wall the defect is bounded away from 1, and a gluing failure is not a limit of large Beck–Chevalley defects. In the L^2 sense nothing fails at all: in finite dimensions $c < 1$, so every compatible pair of L^2 sections glues (theorem 14.6(b)).

Proof. (a) Summing over w gives P_{UV}, and summing over u gives P_{VW}, because both share P_V. The law factorises as $P(u \mid v)P(w \mid v)P(v)$, so $U \perp W \mid V$. Conversely, conditional independence given V forces $P(u, w \mid v) = P(u \mid v)P(w \mid v)$, which with the two windows fixed determines P. Full support makes the meet $\sigma(U, V) \wedge \sigma(V, W)$ equal to $\sigma(V)$, since every u is joined to every w through each v. Then theorem 14.4 identifies Beck–Chevalley with $U \perp W \mid V$. (b) Glue the windows in the removal order, applying (a) at each step with V the overlap of the removed window with the rest; this is Vorob'ev's theorem [15]. For the three pair windows, removing any one leaves an overlap of two singletons $\{X_i\}, \{X_j\}$ that no single remaining window contains, so no removal order exists. (c) At $\varepsilon = \frac{1}{3}$, $a = 0$, so $|s| \le \min(a, b) = 0$ and the fibre is $\{J\}$. Proposition 16.2 gives $c = |2\varepsilon - 1|/(2\varepsilon)$ at J, which is $\frac{1}{2}$ at $\varepsilon = \frac{1}{3}$: given $X_1 = 0$ the table of (X_0, X_2) is $\left(\begin{smallmatrix} 0 & b \\ b & b \end{smallmatrix}\right)$, with correlation $-\frac{1}{2}$, and likewise given $X_1 = 1$ (the formula is check L4b). In finite dimensions the Friedrichs angle between $L^2(A) \ominus L^2(C)$ and $L^2(B) \ominus L^2(C)$ is positive, so $c < 1$. □

16.6 What tropicalisation sees

Proposition 16.4 (Tropical shadow of hallucination)**.** *Over a real non-Archimedean field:*

(a) valuation commutes with marginalisation, so if the tropical tables have no tropical joint, the classical tables have no joint;

(b) the hallucination at constant $\varepsilon < \frac{1}{3}$ is invisible tropically: all entries have valuation 0, and the zero tropical joint exists;

(c) a scaled family with agreement $\varepsilon = \kappa t^\alpha$, $\alpha > 0$, is tropically obstructed. The certificate is combinatorial: every outcome has an agreeing pair, so every tropical joint entry is at least α, while the disagreement cells require 0.

Proof. (a) Marginalisation is pushforward, and theorem 11.3 gives $\mathrm{val}(\mathsf{D}f(P)) = \mathsf{D}_{\mathbb{T}} f(\mathrm{val}\, P)$. (b) and (c) are direct (check L8). □

So the tropical shadow of a hallucination certificate is a support-level certificate: strong contextuality in the sense of [4], graded by rates. It is sufficient but not necessary. The boundary $\varepsilon = \frac{1}{3}$ concerns leading coefficients, which the valuation discards. Hallucination is therefore *not* in general a tropical wall crossing. The wall that detects it is classical and linear in the weights (fig. 8). The tropical walls of proposition 12.2 detect only obstructions that are visible in rates.

Several agents. When several agents each report a tower about the same state, the same tools answer practical questions. Convex order of the towers is Blackwell's information order; two overlapping pairs of logs always reconcile, while three can fail with a cycle certificate (the agent version of proposition 16.3); and a pooling rule beats the better agent iff it lies in a KL ball around the Bayes pool whose radius is the conditional information the other agent adds, which vanishes exactly under Beck–Chevalley. These results, with a bound on decision errors, are in the companion note [18].

17 Places: the context presheaf over $\mathcal{M}(\mathbb{Z})$

The data of section 16 are rational, so the context presheaf can be read at every place of $\mathbb{Q}$ at once. The setting that holds all the places together is the Berkovich spectrum $\mathcal{M}(\mathbb{Z})$ [22, Ex. 1.4.1], studied in detail by Poineau [13]. This section shows that the places separate the gluing problem

by the *kind* of condition involved, not by context. The contexts stay in the lattice; the places sort the linear, positive and valuative parts of the problem. The sheaf-theoretic reading of contextuality is due to Abramsky and Brandenburger [4], and the underlying marginal problem goes back to Vorob'ev [15]. The new points here are the place-by-place decomposition and its consequences for the Beck–Chevalley defect.

The space. $\mathcal{M}(\mathbb{Z})$ is the set of multiplicative seminorms on $\mathbb{Z}$ bounded by the usual absolute value. By Ostrowski's theorem it is a tree with one centre:

- the trivial seminorm a_0, whose residue field is $\mathbb{Q}$ with the trivial absolute value; by Poineau's description the germs of analytic functions at a_0 form the field $\mathbb{Q}$;
- for each prime p a branch $\{|\cdot|_p^{\alpha} : \alpha \in (0,\infty)\}$, whose points have completed residue field $\mathbb{Q}_p$, ending at the seminorm with kernel $p\mathbb{Z}$ and residue field $\mathbb{F}_p$;
- one archimedean branch $\{|\cdot|_\infty^{\varepsilon} : \varepsilon \in (0,1]\}$.

Every neighbourhood of a_0 contains all but finitely many branches in full. A tree is contractible.

Throughout, $J + s\chi$ is the family of proposition 16.1: $a = (3\varepsilon-1)/4$, $b = (1-\varepsilon)/4$, $J(000) = J(111) = a$, $J = b$ elsewhere, and $\chi(x) = (-1)^{x_0+x_1+x_2}$.

Proposition 17.1 (The centre: signed sections always exist)**.** *For every $\varepsilon \in \mathbb{Q} \cap (0,1)$ and every field $K \supseteq \mathbb{Q}$, the signed measures on $\{0,1\}^3$ over K whose three pair tables equal the window data form the line $\{J + s\chi : s \in K\}$. In particular the window data glue over $\mathbb{Q}$, and over every $\mathbb{Q}_p$, for all ε, including $\varepsilon < \frac{1}{3}$, where $a < 0$.*

Proof. J has total mass 1 and the right pair tables by direct computation. The pair-marginal map $K^8 \to K^{12}$ has rank 7: in the character basis $\{(-1)^{\langle u,x\rangle}\}$ the pair tables determine every Fourier coefficient with $|u| \le 2$, and the only character they do not see is χ, the one with $u = (1,1,1)$. So the kernel is $K\chi$. Checks M1 and M2 verify both statements exactly; M2 also finds rank 7 over $\mathbb{F}_p$ for $p = 2, 3, 5, 7, 11$. □

Proposition 17.2 (Positivity is archimedean)**.** *The positive fibre $\{s : J + s\chi \ge 0\}$ is non-empty iff $\varepsilon \ge \frac{1}{3}$. The order used here is the unique ordering of $\mathbb{Q}$, which is the one induced by the archimedean place. No $\mathbb{Q}_p$ admits an ordering, because -1 is a sum of squares in $\mathbb{Q}_p$. Hence the obstruction of proposition 16.1, with its certificate $\Phi = \sum P(X_i \ne X_j) \le 2$, is supported at the archimedean place: at a_0 and along every finite branch the data glue (proposition 17.1).*

Proof. The first statement is $\min(a,b) \ge 0 \iff \varepsilon \ge \frac{1}{3}$. A field is orderable iff -1 is not a sum of squares (Artin–Schreier). In $\mathbb{Q}_p$ every quadratic form in at least five variables is isotropic [16, Ch. IV, Thm. 6], so $x_1^2 + \cdots + x_4^2 + x_5^2$ has a non-trivial zero, and -1 is a sum of four squares. Check M3 confirms the first statement on a grid and exhibits explicit witnesses for every $p \le 47$, lifted by Hensel's lemma to precision p^{12} (for $p = 2$ through $\sqrt{-7} \in \mathbb{Z}_2$). □

Proposition 17.3 (Beck–Chevalley at the centre, its size at the places)**.** *For the windows $\sigma(X_0, X_1)$ and $\sigma(X_1, X_2)$, let Δ_{x_1} be the determinant of the 2×2 table of (X_0, X_2) given $X_1 = x_1$. On the family $J + s\chi$,*

$$\Delta_0 = \tfrac{1}{8}\big(4s - (2\varepsilon-1)(\varepsilon-1)\big), \qquad \Delta_1 = -\tfrac{1}{8}\big(4s + (2\varepsilon-1)(\varepsilon-1)\big).$$

(a) Beck–Chevalley holds iff $\Delta_0 = \Delta_1 = 0$, a polynomial condition with rational coefficients. On the family it holds only at $(\varepsilon, s) = (\frac{1}{2}, 0)$ and at the degenerate point $(1, 0)$. Whether it holds is decided at a_0.

(b) *The size of the defect is archimedean:* $c = \max_{x_1} |\Delta_{x_1}|/\sqrt{r_0 r_1 c_0 c_1}$, *the largest conditional correlation. On the symmetric family row and column margins agree, so* c *is rational, and* $c = |2\varepsilon - 1|/(2\varepsilon)$ *at* $s = 0$. *For a general table* c^2 *is rational and* c *lies in a quadratic field, so for algebraic data one works over* $\mathcal{M}(\mathcal{O}_K)$, *whose branches are the places of* K.

(c) *At a finite place the defect is measured by* $v_p(\Delta_{x_1})$. *By the product formula* $|\Delta|_\infty = \prod_p p^{\,v_p(\Delta)}$ *for* $\Delta \in \mathbb{Q}^\times$, *so the finite places together determine the archimedean size of the determinant, though not the normalised* c.

Proof. The determinants are direct expansions; (a) is theorem 14.4 in coordinates, and the common zeros are found by elimination (the resultant in s is $-(\varepsilon - 1)(2\varepsilon - 1)/8$). (b) is proposition 16.2 with the margins written out. (c) is the product formula. Checks M4–M6. □

At $\varepsilon = \frac{2}{5}$, $s = 0$: $a = \frac{1}{20}$, $b = \frac{3}{20}$, $\Delta_0 = -\frac{3}{200}$ and $c = \frac{1}{4}$. The valuations are $v_2(\Delta_0) = -3$, $v_3(\Delta_0) = 1$, $v_5(\Delta_0) = -2$, $v_7(\Delta_0) = 0$. At the end of the 3-branch, J reduces to a section over $\mathbb{F}_3$ supported on $\{000, 111\}$, because $3 \mid b$; at the end of the 7-branch all eight atoms survive; J is not 2- or 5-integral (check M7). Figure 9 draws the example on the tree.

Remark 17.4 (Why contexts are not separated by primes)**.** The two windows of the Beck–Chevalley square meet only in $\sigma(X_1)$; their nerve is an edge, and the square can be drawn on two branches through a_0, in the same way that the star space of section 4 has one centre and many leaves. The three pair windows of proposition 16.1 have a hollow triangle as nerve ($b_0 = b_1 = 1$), and a tree cannot carry that cycle. Moreover, by weak approximation the valuations of one rational law can be prescribed independently at finitely many primes, so assigning contexts to primes imposes no gluing condition between them. Both facts point the same way as proposition 17.2: the obstruction lives in the coefficients (positivity at ∞), not in the topology of $\mathcal{M}(\mathbb{Z})$. Check M8 computes the Betti numbers.

Remark 17.5 (Positive tropicalisation and the archimedean branch)**.** The positive tropicalisation of section 11 needs a valuation and an order at once. Over a real closed non-Archimedean field the order restricts to the unique ordering of $\mathbb{Q}$, which belongs to the archimedean place. So the tropical layer is not one of the p-adic branches: p-adic valuations are blind to sign ($v_p(-x) = v_p(x)$) and allow cancellation of positive quantities ($v_3(\frac{1}{3}) = v_3(\frac{2}{3}) = -1$, yet the sum is 1). It reads naturally as the degeneration $\varepsilon \to 0$ of the archimedean branch, in the sense of the hybrid spaces of Boucksom and Jonsson [14]. We have not proved this (open problem 17.6).

Open problem 17.6 (The tropical layer as a hybrid degeneration)**.** Realise the positive tropicalisation of section 11 as the restriction of the context presheaf to the end of the archimedean branch in a hybrid analytic space over $\mathbb{Z}$, compatibly with propositions 17.1 to 17.3.

An interactive view. Figure 10 shows the explorer that accompanies this section and section 16 (`explorer/context_sheaf.html`, a single page with no dependencies). It puts the six layers of the example side by side: the contexts and their window tables, the marginal polytope, the fibre of global laws, the conditional-expectation flow with the defect c and $I(X_0; X_2 \mid X_1)$, the support face, and the outcome. Check E1 reproduces its numbers at $\varepsilon = 0.4$.

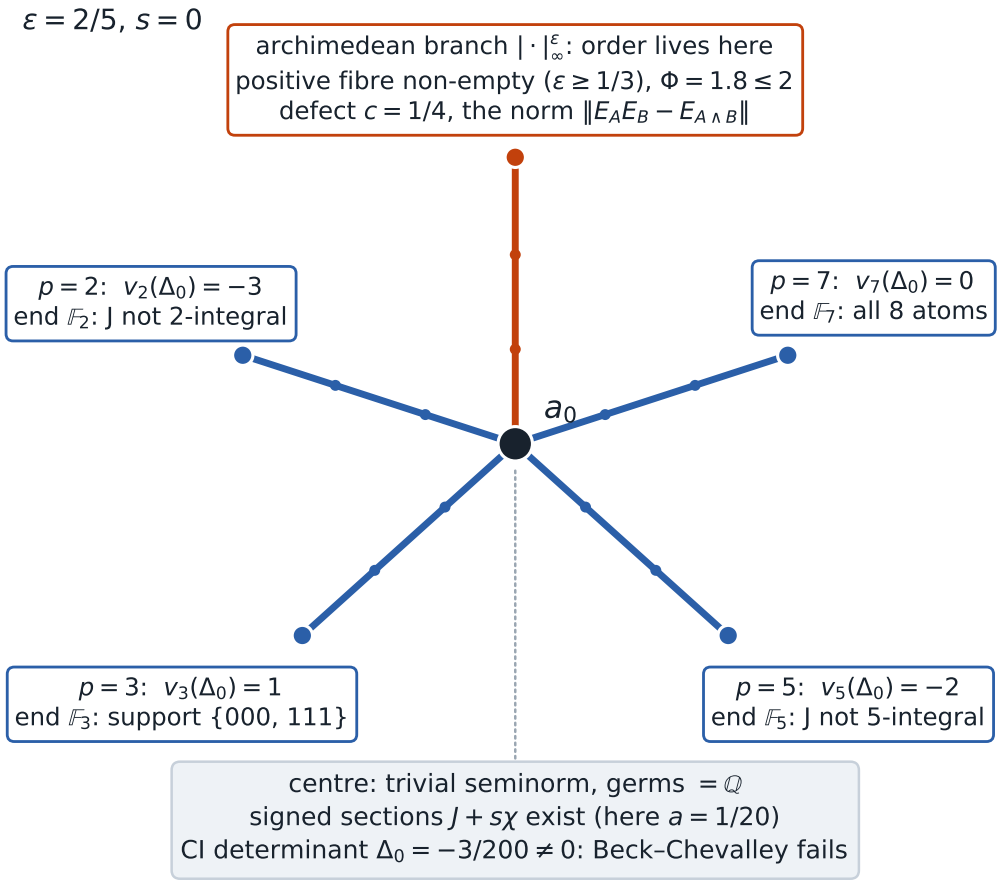


Figure 9: The three-token context presheaf at $\varepsilon = \frac{2}{5}$, read at the places of $\mathbb{Q}$ on the tree $\mathcal{M}(\mathbb{Z})$. The centre sees the linear gluing and decides Beck–Chevalley; each finite branch sees valuations ($|\cdot|_p^\alpha$ along the branch) and, at its end, the reduction mod p; only the archimedean branch carries order, positivity and the normalised defect c. Small dots on a branch mark points $|\cdot|_p^\alpha$ for $\alpha \in (0, \infty)$. Generated by `atlas/fig_places.py`.



Figure 10: The explorer at $\varepsilon = 0.40$, $s = 0$: liftable, with $\Phi = 1.8$, $c = 0.25$ and $I(X_0; X_2 \mid X_1) = 0.0464$ bits. The phase line shows the fibre length $2\min(a, b)$ and the smallest defect in the fibre, which is 0 only at $\varepsilon = \frac{1}{2}$.

18 Empirical tests of the hallucination example

The tests were pre-registered before any of the statistics below was computed, in the file `PREREG_hallucination.md` of the folder `halluc_study` (its SHA-256 begins `4a34e1bb`). The same file discloses an earlier, different analysis of the same data in the validation paper. There were three studies. Study 1 runs on public data. Study 2 was run on samples from a local model; its analysis follows a frozen amendment. Study 3 needs pairwise elicitation from a model and ships as a kit with a synthetic smoke test.

18.1 Why only the second layer can be tested on existing data

The obstruction of proposition 16.1 needs each pair of tokens to be observed in a separate context. In existing benchmarks every item's answers are observed jointly, and a jointly observed table always has a joint law, so the gap below $\frac{1}{3}$ cannot appear in them. Study 3 is designed to collect pairwise data. On jointly observed data the testable layer is the second one: the window Beck–Chevalley defect of proposition 16.2.

18.2 Study 1: LLM-drift answers

Data. The data are the answers of GPT-3.5 (0301, 0613) and GPT-4 (0314, 0613) to the same items [5]: four runs per item, one answer each. Five tasks have a reference answer comparable by equality: USMLE (intact prompts), primality, happy numbers, HotpotQA (agent) and ARC. The extraction rules are those of the validation paper, unchanged.

P1: consistency detector. For a target run, the score is the fraction of the other runs whose valid answer differs from the target's. The outcome is whether the target answer is wrong. This is the cross-sample consistency principle of SelfCheckGPT [9], with the other runs playing the role of samples.

P2: window defect. The binary correctness of GPT-4 March, GPT-4 June and GPT-3.5 June is read through two windows:

$$A = \sigma(Y_{4\mathrm{M}}, Y_{4\mathrm{J}}) \text{ (same model, two dates)}, \qquad B = \sigma(Y_{4\mathrm{J}}, Y_{35\mathrm{J}}) \text{ (same date, two models)}.$$

They share $Y_{4\mathrm{J}}$. By proposition 16.2, Beck–Chevalley holds iff the March GPT-4 errors and the June GPT-3.5 errors are independent given the June GPT-4 error. The defect c is tested against a parametric Markov null.

Table 1: Study 1 (pre-registered). AUROC of the disagreement score for target runs GPT-4 June (primary) and GPT-3.5 June (secondary), and the window Beck–Chevalley defect c with its Markov-null median and Holm-adjusted p-value ($B = 2000$; 95% bootstrap intervals).

task	n	wrong	AUROC (4J)	AUROC (35J)	n_c	c	null	p_{Holm}
USMLE	340	52	0.692 [0.617, 0.766]	0.895 [0.853, 0.930]	338	0.096 [0.033, 0.401]	0.106	0.756
primes	1000	489	0.881 [0.861, 0.901]	0.739 [0.704, 0.771]	949	0.498 [0.411, 0.584]	0.047	0.003
happy numbers	500	324	0.873 [0.840, 0.902]	0.888 [0.855, 0.917]	387	0.098 [0.020, 0.211]	0.082	0.756
HotpotQA	4844	2693	0.708 [0.694, 0.721]	0.800 [0.783, 0.817]	324	0.309 [0.128, 0.519]	0.083	0.005
ARC	467	341	0.821 [0.779, 0.862]	0.902 [0.854, 0.945]	467	0.320 [0.200, 0.446]	0.069	0.003

Results.

- *H1 is supported.* Disagreement with the other runs detects wrong GPT-4 June answers in all five tasks, with AUROC between 0.69 and 0.88 and every interval above 0.5.
- *H2 is supported.* Beck–Chevalley fails in three tasks: primes ($c = 0.50$ against a null median of 0.05), HotpotQA (0.31) and ARC (0.32). In these tasks the errors of the two outer runs stay coupled after conditioning on the shared run. By proposition 16.2, this is coupling that neither window holds: a shared item-level difficulty that a single run does not screen off. In USMLE and happy numbers the defect is indistinguishable from the Markov null.
- *Exploratory.* The rank correlation between c and the GPT-4 June AUROC across the five tasks is 0.70. With five tasks this carries no inferential weight.

What Study 1 does and does not show. It shows that the window defect is measurable on real model outputs, and that it is nonzero in several tasks. It also replicates the consistency principle that underlies black-box hallucination detection. It does not test the obstruction below $\frac{1}{3}$, for the reason given above. Its errors are factual errors against a reference, a common proxy for hallucination [6] but not the same as an obstructed claim in the sense of [1, §10.5].

18.3 Study 2: HaluEval QA

Data and amendment. HaluEval QA [7] (MIT licence) gives, for each item, knowledge, a question, a right answer and a hallucinated answer. Llama 3.1 (8B, run locally through Ollama) was sampled $K = 5$ times per prompt at temperature 1 on items 0–1999, with the system line "Answer with a short phrase only". Items 0–199 were inspected first; that inspection found that answer length alone separates the two answer types, and that the containment judge misses paraphrases. Amendment 1 (`AMENDMENT1.md`, SHA-256 beginning `07782f4e`) was frozen after that and before any statistic on items 200–1999 was computed. It adds a length baseline, a length-adjusted AUROC, a logistic test of the detector beyond length, a relaxed deterministic judge, and a within-sample window test. Deviations from the original registration: $N = 2000$ rather than 10 000, the system line, and the choice of model. The confirmatory set is items 200–1999 ($n = 1800$).

Table 2: Study 2, confirmatory set ($n = 1800$ items, 3600 answers; Amendment 1). Detector = fraction of the five samples that do not match the answer. Length-adjusted AUROC compares answers only within the same length quintile. LR = likelihood-ratio statistic for adding the detector to log(length) in a logistic model. 95% bootstrap intervals over items ($B = 2000$).

judge	AUROC	length-adjusted AUROC	LR (p)
containment (pre-registered)	0.871 [0.860, 0.881]	0.851 [0.827, 0.874]	404 ($< 10^{-80}$)
relaxed (Amendment 1)	0.870 [0.859, 0.882]	0.883 [0.861, 0.903]	493 ($< 10^{-100}$)
length alone	0.974	—	—

Results.

- *The detector works, but length is the stronger cue in this benchmark.* The pre-registered AUROC is 0.871. Answer length alone reaches 0.974: HaluEval's hallucinated answers are usually longer than the right ones. So the raw AUROC cannot be read as evidence for the detector.

- *H1″ is supported: the detector adds information beyond length.* Within length quintiles the detector still separates the two answer types (AUROC 0.85 with the containment judge, 0.88 with the relaxed judge), and adding it to a length-only logistic model improves the fit by LR = 404 (cluster-robust Wald $p < 10^{-80}$, allowing for the two answers per item). The adjustment is uneven: the middle quintile holds most of the comparisons, and in the outer quintiles one answer type has only 7 to 31 members.

- *H2′ is supported: the within-sample windows fail Beck–Chevalley.* Let $Y_k = 1$ if sample k misses the right answer (relaxed judge; 14.7% of samples). With windows $\sigma(Y_1, Y_2)$ and $\sigma(Y_2, Y_3)$, the mass-weighted defect is $\bar{c} = 0.32$ against a permutation-null median of 0.03 ($p = 0.0005$). Adding the difficulty count $D = Y_4 + Y_5$ to the centre reduces it to 0.13 (null 0.02, $p = 0.005$).

Interpretation. The five samples are independent given the item, so the defect does not come from the sampling. It comes from a variable missing from the lattice: the item itself. The wrong-count distribution shows this. 1441 items have no wrong sample and 183 have all five wrong, against 813 and 0 expected under one common error rate. A model that is wrong on an item tends to be wrong in the same way every time. One sample in the centre does not screen this off; adding two more absorbs most of it, as the centre-enlargement reading of proposition 16.2 predicts. This is a common cause, not an obstruction: the samples are jointly observed, so a joint law exists. The 183 consistently wrong items are where a consistency detector is blind.

Does the test detect Beck–Chevalley correctly? The theorem side is exact: Beck–Chevalley holds iff the outer variables are independent given the centre (theorem 14.4, checks H1, H2, L4). The finite-sample test was checked separately on synthetic data (`check_bc_test.py`).

- *When Beck–Chevalley holds it rejects at the nominal rate.* On random Markov chains $Y_1 - Y_2 - Y_3$ the rejection rate at level 0.05 is 0.056 ($n = 300$) and 0.060 ($n = 1800$) over 1000 replications, with uniform p-values. The population defect is 0 for all of 200 random Markov laws.

- *When it fails it detects it.* With item-level latent difficulty (samples i.i.d. given the item) the population defect is 0.35. The power is 0.76 at $n = 100$ and 1.00 from $n = 300$, and the sample $\bar{c}$ matches the population value at every n.

- *Centre enlargement works as proposition 16.2 says.* With the latent class in the centre, Beck–Chevalley holds again and the rejection rate is 0.042. The proxy $D = Y_4 + Y_5$ only shrinks the defect ($0.35 \to 0.10$).

- *Replica of Study 2.* A three-class binomial model fitted to the wrong-count distribution alone gives simulated $\bar{c}$ intervals $[0.30, 0.56]$ (centre Y_2) and $[0.07, 0.41]$ (centre (Y_2, D)). Both contain the observed values 0.32 and 0.13. This model has as many parameters as free cells, so the replica shows consistency, not a test of the latent reading.

Limitations. Both judges are string-based. The relaxed judge still misses name variants ("Don Henley" against "Donald Hugh Henley"), which inflates the detector score of some right answers. A meaning-based judge is the next step. HaluEval's hallucinated answers were written to a template, which produces the length gap; the length-adjusted results are the ones to cite.

18.4 Study 3: kit

- *Study 3* (`kit_pairwise_cycles.py`) asks about each pair of three related claims in a separate prompt, and evaluates the 3-cycle inequalities of [2, Thm. 4.8]. It reports the largest violation V with a bootstrap interval. For the symmetric model $V = 2(1-3\varepsilon)$; the smoke test recovers $V \approx 0.5$ at $\varepsilon = 0.25$ and no violation at $\varepsilon = 0.45$. This is the design in which a hallucination in the sense of proposition 16.1 can be observed directly.

TruthfulQA [8] can supply claim triples for Study 3 once model answers are collected.

19 Computational checks

19.1 The scripts

- `check_shadows.py` has 30 checks, in exact rational arithmetic except A1(v) and A4, which use double precision with tolerance 10^{-9}. It imports the tower objects from `check_tower.py`.
- `check_tropical.py` has 8 checks, in exact Puiseux arithmetic except the floating-point ranks in T6 and T7.
- `check_cohomology.py` has 10 checks, exact except H2.
- `check_hallucination.py` has 9 checks, exact except L4b, L5 and L7.
- `check_infogeom.py` has 5 checks (I1–I5) of section 14.2.
- `check_twocat.py` has 12 checks (TC1–TC12) of lemmas 6.12 and 6.13 and propositions 6.14, 6.16, 6.17 and 6.20; it runs the tower code's own C_3, δ and $\bar{\lambda}$.
- `check_barr.py` has 4 checks (R1–R4) of lemma 6.11, by linear-programming feasibility on random instances, with negative controls and a counterexample to the converse of (d).
- `check_places.py` has 9 checks (M1–M8, E1), exact except the floating-point I in E1.
- `check_bc_test.py` has 7 Monte Carlo checks (S1–S6) of the finite-sample Beck–Chevalley test of section 18.
- In the folder `halluc_study`, `study1_llmdrift.py` runs the pre-registered Study 1 (table 1), and `kit_halueval.py` and `kit_pairwise_cycles.py` are the kits for Studies 2 and 3, with smoke tests.
- `atlas/fig_places.py` draws fig. 9; `explorer/context_sheaf.html` is the explorer of fig. 10.
- `atlas/atlas.py` draws figs. 1 to 8, and `atlas/fibre_scan.py` computes the Betti map. Both use `atlas/fibre_topology.py`.

All checks pass; the hashes are in `FROZEN.txt`.

A1–A7b	theorem 3.3 (449 commuting, 151 non-commuting); proposition 3.4; example 3.5; theorem 3.6; corollary 3.8; lemma 3.2; proposition 3.9
B1–B7	star space; theorem 4.9; proposition 4.10; theorem 5.3; proposition 4.11; propositions 5.1 and 5.2; proposition 5.4
C1–C7	audit of earlier scripts; proposition 4.2; example 4.3; proposition 5.5(b)
D1–D2, E1–E2	proposition 12.3; theorem 9.4
G1–G3	proposition 10.1; proposition 4.6; proposition 6.8
T1–T8	theorem 11.3; theorem 11.6; theorem 11.11(a); proposition 12.1; example 11.10; proposition 12.2(a)
H1–H5	theorem 14.4; theorem 14.6(b, d); proposition 14.7
K1–K5	theorems 15.3 and 15.5; proposition 15.4(b); fig. 7
L1–L8	propositions 16.1, 16.2 and 16.4; the token stream of fig. 8
R1–R4	lemma 6.11
TC1–TC12	lemmas 6.12 and 6.13 and propositions 6.14, 6.16, 6.17 and 6.20; example 6.15 and theorem 6.18
I1–I5	propositions 14.8 and 14.11 and theorem 14.9; signed versus positive sections
M1–M8, E1	propositions 17.1 to 17.3; remark 17.4; figs. 9 and 10
S1–S6	the permutation test of section 18: size, power, centre enlargement, replica of table 2

19.2 What checks establish

Finite exact checks verify identities on the enumerated inputs and can falsify overbroad claims. They do not prove identities for all measures, establish measurability, or supply categorical typing. Every general statement in this paper has a proof; the checks guard the proofs against slips. Hashes certify the identity of artefacts, not the truth of theorems.

19.3 Audit of earlier scripts

Five scripts accompanied an earlier version. All five share one nearest-atom `pooling(nu, atoms)`, with "type-aware" total variation and ties broken by key order. They pass every reported check for the following reasons:

Script	Why it passes
`check_giry.py`	Π always has two atoms, and Σ's descriptors i mod 2 biject with them. So $\sigma(\theta) = \sigma(\tilde{r})$, and L and R have identical cells (C1).
`check_3cat-Coherence.py`	As above. In addition, level-5 atoms of type D^2 against inputs of type D^1 are all at TV distance 1, so the pooling is total, $e_\perp$, and the "3-cell" is trivial (C7).
`check_non-coherent.py`	The duplicated predictive $(0, \rho_1), (1, \rho_1)$ ties, and the first key wins, so R's cells equal L's (C6).
`double_category.py`	"Interchange" is $R(L\nu) = L(R\nu)$ with $L = R$. "(U) defect nonzero" is $e_\perp \neq e_\top$.
`check_span.py`	The cells coincide (C1). `dist_of` strips θ, so "not inverse" holds by a change of key type. "(Cm) equals sum" compares a pair with itself (C5). Mixed inputs combine two key types under one metric.

By proposition 4.2, any composite of nearest-atom poolings is one pooling onto a coarsening of the first partition. By proposition 12.1, such pooling is a zero-temperature shadow. The scripts' counts therefore carry no information about theorem 3.3. A meaningful test uses the star space with labels that are not functions of the predictive, and compares three pairs: canonical shadows, label-only shadows, and naive composites. Checks B2, B3 and C2 do exactly this.

20 Consequence for the tower paper

The tower paper [1] never uses $E_B E_A = E_{A \vee B}$. Its poolings live on a fixed sample index [1, Def. 13.13], and its order in [1, Thm. 13.16] is a transitive closure. An unnumbered paragraph added after that theorem explains why the closure is needed. A two-step label pooling $E_{G_2} E_{G_1}$ is one conditional expectation iff the two commute, which by theorem 3.3 means conditional independence given $G_1 \wedge G_2$; the product is then $E_{G_1 \wedge G_2}$, never the join.

21 Summary and open problems

- Two conditional expectations commute iff they are conditionally independent given their meet. Otherwise they interchange laxly, through the meet, at the exact rate c^{2n-1}. The join identity is false except for equal σ-algebras.
- The canonical shadows of a context live on its star space. They form a commutative band, are natural along context maps that are injective on predictive atoms, and interchange strictly.
- The tower acts on barycentre fibres as a martingale. $e_\perp$ and $e_\top$ are the bottom and top of each fibre. Each computed Beck defect is one conditional-expectation step.
- No coupling lifting of Barr or martingale type makes the averaging law a lax mixed distributive law: label-preserving orders keep the comultiplication monotone but cannot orient its cell, and the order that orients the cell breaks the counit. In the 2-category whose 2-cells are couplings the cell exists, and the obstruction moves to whiskering, which preserves composites only up to a comparison measured by the Beck–Chevalley defect.
- Over a real non-Archimedean field the whole picture tropicalises exactly: the valuation is a monad morphism, fibres lift, mixtures become tropical secants, and the valuation factors through Berkovich spaces. Hard assignment is the zero-temperature shadow.
- Conditional expectation is restriction in the L^2 presheaf. Commutation is its Beck–Chevalley condition, and gluing fails only when $c = 1$. Tropical fibres have computable cohomology, read off from certificates of impossible assignment patterns. In the Hardy–Weinberg example the collision wall is a jump $\Delta b_0 = -1$.
- In a three-token model a hallucination is a failure to glue pairwise windows, certified by a Bell-type functional, below a straight wall in source weights. Two windows always glue, so failure needs a cycle in the cover; the Beck–Chevalley defect is a separate layer and stays at $\frac{1}{2}$ at the wall. Above the wall, overlapping windows commute iff the tokens are Markov through the shared token, and the Friedrichs cosine is the exact measure of the coupling that no window holds.

- For several agents (companion note [18]): Blackwell order of the towers, reconciliation of pairwise logs, and the pooling rules that beat the better agent, which form a KL ball whose radius vanishes exactly under Beck–Chevalley.
- Over $\mathcal{M}(\mathbb{Z})$ the gluing problem splits by place. The linear part glues everywhere, Beck–Chevalley is decided at the trivial seminorm, finite places measure the defect by valuations, and positivity, hence the hallucination obstruction, lives only at the archimedean place.

The open problems are open problems 6.21, 8.1, 12.5 to 12.7 and 17.6, and conjecture 12.4.

A Changes from the previous versions

This paper supersedes three manuscripts of October 2026:

(i) *Conditional Expectations, Beck Defects, and a Proposed Double-Category Structure*, referred to below as G;

(ii) *Conditional Expectations, Beck Defects, and the Shadow Algebra of the Posterior Tower*, referred to below as S;

(iii) *From Barycentre Fibres to Secant Geometry.*

S and (iii) are absorbed in full. The table lists every statement of G that changed.

G	Status	Here
Def 3.2, Props 3.3, 3.5 (label shadows)	kept	proposition 4.1; composites in proposition 4.2
Thm 4.2 (nested)	kept	special case of theorem 3.3
Thm 4.4 (CI $\Rightarrow$ commute)	strengthened	five-way equivalence, theorem 3.3
Rem 4.5 (join), Ex 4.6	kept, sharpened	proposition 3.4; example 3.5
Rem 4.7 ("hypotheses necessary")	made precise	theorem 3.3; general case theorem 3.6 and corollary 3.8
Prop 4.8, Rem 4.9, Prop 8.1 (inheritance)	discharged	proposition 4.5
Defs 5.1, 5.3, Rem 5.4, Defs 5.5, 5.6, Props 5.7, 5.8, Rem 5.9	kept	section 6; context maps defined concretely in definition 6.1
Def 5.10, Prop 5.11 (compatible subcategory)	proved	propositions 4.6 and 6.8
Diagram after Def 5.13	withdrawn	horizontal composition of contexts is not a functor $\mathbf{Ctx} \times \mathbf{Ctx} \to \mathbf{Ctx}$
Thm 5.14 (interchange blueprint)	replaced	proposition 6.10
Prop 6.1 (required data)	kept	proposition 7.1, now supplied by proposition 7.2
Conj 6.2 (two-shadow factorisation)	superseded	one step suffices in every computed case

G	Status	Here
Thms 7.1, 7.2 (Options A, B)	partly invalid	proposition 7.4: (Cm) does not decompose
§8 (Giry)	completed	section 8
Prop 9.1, Rem 9.2 (3-cells)	kept	triviality in quintets
§10 (regimes)	kept	worked regimes in section 3
Def 11.1, Thms 11.2, 11.5 (double category of the tower)	withdrawn	2-cells mistyped; proposition 6.10
Thm 11.4, Prop 11.8	kept	identities and latent relabellings are natural (section 10)
Prop 11.7 (Gaussian interchange)	false	proposition 3.9
§11.6.1 (restricted domain V)	corrected	the square is total (section 10)
§12 (scope of checks)	kept	section 19
§13 (Berkovich programme)	replaced	section 11
§14 (stochastic-to-convex bridge)	kept, strengthened	theorem 9.4: fully faithful; shadows not affine
§15 (Wasserstein–Berkovich functor)	withdrawn	ill-defined; theorem 11.11 instead
§16 (secants, wall-crossing)	corrected	section 12: order ≥ 3; hierarchy invisible to moments; conjecture 12.4

Code and data availability

All scripts, figure generators, the interactive explorer, the pre-registration and its amendment, and the study kits are at `https://github.com/vkawasth/Trainingless-Transformer-NoGradientDescent/tree/main/papers`. The HaluEval and LLM-drift data are downloaded from their original sources; the Study 2 samples (model outputs on HaluEval prompts) are included.

Acknowledgements

We thank the people who collected and released the data and software used here:

- L. Chen, M. Zaharia and J. Zou for the LLM-drift answers [5];
- J. Li, X. Cheng, W. X. Zhao, J.-Y. Nie and J.-R. Wen (RUCAIBox) for HaluEval [7], released under the MIT licence;
- S. Lin, J. Hilton and O. Evans for TruthfulQA [8];
- the developers of Python, NumPy, SciPy (including the HiGHS solver), SymPy, pandas and Matplotlib, on which every computation and figure depends.

The data are used under their licences and terms; no credentialed or patient-level data were used. The study in section 18 was pre-registered and frozen by hash before its statistics were computed. Independent reviews by language-model referees found errors in earlier drafts of this material,

which are corrected here, and an AI assistant (Claude, Anthropic) helped draft text, write the exact-arithmetic checks and produce the figures. The author is responsible for all remaining errors.

References

[1] V. K. Awasthi, The three-level posterior tower: hierarchical Bayesian inference and mixed distributive laws. Technical report, October 2026 (version 10).

[2] V. K. Awasthi, Conditionals, couplings and toric geometry. Technical report, October 2026.

[3] V. K. Awasthi, Pooling idempotents and the Beck defects of the absolute tower. Note, October 2026.

[4] S. Abramsky and A. Brandenburger, The sheaf-theoretic structure of non-locality and contextuality. *New J. Phys.* **13** (2011), 113036.

[5] L. Chen, M. Zaharia and J. Zou, How is ChatGPT's behavior changing over time? *Harvard Data Science Review* **6** (2024), no. 2; arXiv:2307.09009. Data: `https://github.com/lchen001/LLMDrift`.

[6] Z. Ji, N. Lee, R. Frieske, T. Yu, D. Su, Y. Xu, E. Ishii, Y. J. Bang, A. Madotto and P. Fung, Survey of hallucination in natural language generation. *ACM Comput. Surv.* **55** (2023), no. 12, Article 248.

[7] J. Li, X. Cheng, W. X. Zhao, J.-Y. Nie and J.-R. Wen, HaluEval: a large-scale hallucination evaluation benchmark for large language models. In *Proc. EMNLP* (2023), 6449–6464. Data: `https://github.com/RUCAIBox/HaluEval`.

[8] S. Lin, J. Hilton and O. Evans, TruthfulQA: measuring how models mimic human falsehoods. In *Proc. ACL* (2022), 3214–3252.

[9] P. Manakul, A. Liusie and M. J. F. Gales, SelfCheckGPT: zero-resource black-box hallucination detection for generative large language models. In *Proc. EMNLP* (2023), 9004–9017.

[10] C. Améndola, J.-C. Faugère and B. Sturmfels, Moment varieties of Gaussian mixtures. *J. Algebr. Stat.* **7** (2016), 14–28.

[11] D. Alessandrini, Logarithmic limit sets of real semi-algebraic sets. *Adv. Geom.* **13** (2013), 155–190.

[12] H. H. Bauschke, F. Deutsch and H. Hundal, Characterizing arbitrarily slow convergence in the method of alternating projections. *Int. Trans. Oper. Res.* **16** (2009), 413–425.

[13] J. Poineau, *La droite de Berkovich sur* $\mathbb{Z}$. Astérisque **334**, Soc. Math. France, 2010.

[14] S. Boucksom and M. Jonsson, Tropical and non-Archimedean limits of degenerating families of volume forms. *J. Éc. polytech. Math.* **4** (2017), 87–139.

[15] N. N. Vorob'ev, Consistent families of measures and their extensions. *Theory Probab. Appl.* **7** (1962), 147–163.

[16] J.-P. Serre, *A Course in Arithmetic*. Graduate Texts in Mathematics 7, Springer, 1973.

[17] M. Barr, Relational algebras. In *Reports of the Midwest Category Seminar IV*, LNM 137, Springer, 1970, 39–55.

[18] V. K. Awasthi, Pooling informed agents: Blackwell order, reconciliation and the KL ball. Companion note, October 2026 (folder `companion`).

[19] S. Amari and H. Nagaoka, *Methods of Information Geometry*. AMS and Oxford University Press, 2000.

[20] I. Csiszár, I-divergence geometry of probability distributions and minimization problems. *Ann. Probab.* **3** (1975), 146–158.

[21] J. Hansen and R. Ghrist, Toward a spectral theory of cellular sheaves. *J. Appl. Comput. Topol.* **3** (2019), 315–358.

[22] V. G. Berkovich, *Spectral Theory and Analytic Geometry over Non-Archimedean Fields*. AMS, 1990.

[23] M. Develin, Tropical secant varieties of linear spaces. *Discrete Comput. Geom.* **35** (2006), 117–129.

[24] M. Develin, F. Santos and B. Sturmfels, On the rank of a tropical matrix. In *Combinatorial and Computational Geometry*, MSRI Publ. 52, 2005, 213–242.

[25] J. Draisma, A tropical approach to secant dimensions. *J. Pure Appl. Algebra* **212** (2008), 349–363.

[26] C. Ehresmann, Catégories structurées. *Ann. Sci. Éc. Norm. Sup.* **80** (1963), 349–426.

[27] M. Giry, A categorical approach to probability theory. LNM 915, Springer, 1982, 68–85.

[28] M. Grandis, *Higher Dimensional Categories: From Double to Multiple Categories*. World Scientific, 2019.

[29] P. Jell, C. Scheiderer and J. Yu, Real tropicalization and analytification of semialgebraic sets. *Int. Math. Res. Not.* (2022), 928–958.

[30] S. Kayalar and H. L. Weinert, Error bounds for the method of alternating projections. *Math. Control Signals Systems* **1** (1988), 43–59.

[31] B. Kulis and M. I. Jordan, Revisiting k-means: new algorithms via Bayesian nonparametrics. *ICML* 2012.

[32] A. Björner, Topological methods. In *Handbook of Combinatorics*, Vol. II, Elsevier, 1995, 1819–1872.

[33] F. Deutsch, *Best Approximation in Inner Product Spaces*. CMS Books in Mathematics, Springer, 2001.

[34] A. Schrijver, *Theory of Linear and Integer Programming*. Wiley, 1986.

[35] S. Payne, Analytification is the limit of all tropicalizations. *Math. Res. Lett.* **16** (2009), 543–556.

[36] R. R. Phelps, *Lectures on Choquet's Theorem*, 2nd ed. LNM 1757, Springer, 2001.

[37] V. Strassen, The existence of probability measures with given marginals. *Ann. Math. Statist.* **36** (1965), 423–439.

[38] J. von Neumann, *Functional Operators, Vol. II*. Princeton University Press, 1950.